

EVOLVING EVALUATION SYSTEMS FROM HUMAN FEEDBACK FOR OPEN-ENDED TASKS

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ABSTRACT

Evaluating open-ended outputs requires task-dependent judgments over multiple aspects of quality that must be balanced in context. Existing proxy metrics and model-based evaluators remain imperfect on such judgments and do not consistently align with human preferences (Fabbri et al., 2021; Hu et al., 2024). Humans can often tell which of two concrete outputs is better, but it is much harder to turn that ability into a comprehensive and effective evaluation procedure (Shankar et al., 2024). So we study evaluator self-evolution by treating the evaluation procedure itself as the object of iterative improvement from human preferences. An evaluation error revealed by human preferences often suggests several plausible ways to improve the evaluator, while limited human feedback provides only weak evidence for deciding which direction to pursue. This ambiguity matters especially in open-ended evaluation, where a useful direction may require several related revisions before producing a measurable gain. An unsuccessful early candidate evaluator therefore leaves the optimizer uncertain whether the direction is wrong or simply underdeveloped. In the default evaluator-evolution setup we tested, this uncertainty often led subsequent revisions toward alternative directions rather than deeper development of the current one. We therefore introduce ERA, a depth-first evolution method that explicitly maintains an improvement direction across revisions and continues developing it until diagnostic results provide a clear reason to move on or its search budget is exhausted. ERA keeps candidate acceptance separate from direction continuation, allowing diagnostic results from unsuccessful candidates to guide further implementation without replacing the best evaluator found so far. Across diverse open-ended tasks, our qualitative analyses show that ERA develops promising directions more thoroughly, produces more sustained improvements, and yields evaluators that better reflect human preferences under matched search budgets. These improvements also transfer downstream: evolved evaluators support better output selection and more effective critic-guided refinement. On representative tasks, using their scores as training rewards further improves generation quality.

1 INTRODUCTION

Evaluating open-ended tasks requires case-specific judgments across multiple dimensions of quality, with no single evaluation standard sufficient for every case (Ye et al., 2024; Kim et al., 2024c). A summary may need to preserve fine-grained details when completeness matters, yet emphasize brevity and the central argument when rapid understanding is the goal. Likewise, visual richness may be valuable for a promotional webpage but detrimental to a task-oriented interface. Metrics and model-based evaluators capture useful aspects of quality, yet do not consistently match human preferences (Fabbri et al., 2021; Xu et al., 2023; Zheng et al., 2023; Hu et al., 2024). Humans themselves, however, often cannot fully express their judgments as a comprehensive, executable evaluation procedure (Shankar et al., 2024). Human preferences are therefore better treated as iterative feedback for improving an evaluator.

Existing approaches use human feedback to train evaluator models (Xu et al., 2023), learn or refine evaluation criteria (Liu et al., 2024; Lv et al., 2026), and select evaluation functions (Shankar et al.,

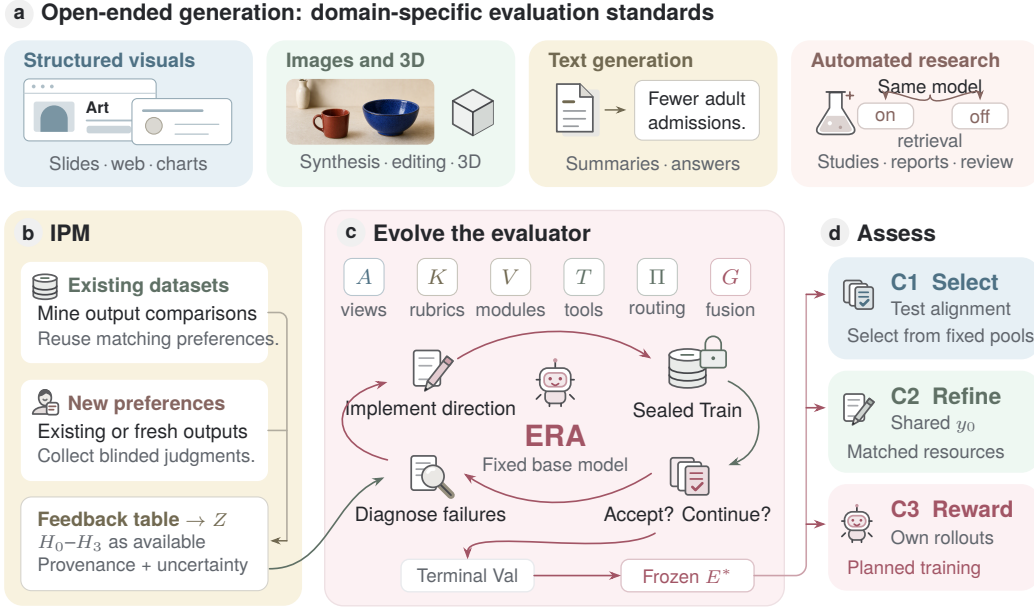


Figure 1: Evaluator self-evolution from human feedback. (a) Four artifact families retain distinct evaluation standards. (b) Informative Preference Mining (IPM) selects comparisons from existing datasets or fresh outputs, reusing matching preferences and collecting missing ones. Both routes retain their provenance and annotation quality. (c) ERA evolves a six-component evaluation procedure with a fixed base model, separating candidate acceptance from direction continuation. (d) C1 alignment and output selection, C2 critic-guided refinement, and planned C3 reward-guided training. Artifacts and feedback are constructed illustrations; the photograph is generated with GPT Image 2. Arrows denote information flow, not measured improvements.

2024). Prompt and skill optimizers offer a complementary mechanism for revising instructions from execution feedback (Agrawal et al., 2026; Yang et al., 2026). We focus on evaluator self-evolution, which directly revises an explicit evaluation procedure while building on the base model’s existing semantic and reasoning capabilities. This allows the evaluator to adapt without updating the base model parameters and makes individual changes easier to inspect and test.

Extending evaluator self-evolution to new open-ended domains, however, raises two challenges: obtaining useful human preferences and reliably turning them into improvements to the evaluation procedure.

Obtaining preference data that is useful for evolution is itself nontrivial: under a limited human feedback budget, the choice of comparisons can affect learning efficiency (Muldrew et al., 2024; Oh et al., 2025). Hand-designed comparisons tend to reflect the cases and failure modes anticipated by the designer, while broad collections can contain many easy or redundant comparisons that reveal little about where the evaluator still falls short. We therefore use Informative Preference Mining (IPM) to prioritize comparisons on which existing evaluation signals disagree. When preference labels already exist, IPM selects informative examples from the available data; otherwise, it prioritizes comparisons for new human annotation.

IPM selects the comparisons used for evaluator evolution, but does not specify how the evaluator should be revised. Pairwise preferences identify the better output without specifying the fine-grained criteria behind that choice (Ye et al., 2024; Lv et al., 2026). The same evaluation error revealed by human preferences can therefore suggest several plausible improvement directions, while limited human feedback often provides little basis for choosing among them. This ambiguity is especially important in open-ended evaluation, where a useful direction may require several related revisions before producing a measurable gain. An unsuccessful early candidate evaluator therefore leaves open whether its direction is ineffective or simply underdeveloped.

This uncertainty also affects how evaluator evolution proceeds. In the default evaluator-evolution setup we tested, each new revision was proposed by the model from previous candidates and their outcomes. We observed a tendency for unsuccessful early candidates to be followed by revisions in alternative directions rather than deeper development of the current one. Over repeated iterations, evolution can therefore spread effort across multiple plausible directions before any one of them has been developed far enough to judge reliably.

We introduce ERA, a depth-first evaluator evolution method that explicitly maintains an improvement direction across revisions (Figure 1c). ERA continues developing the current direction after unsuccessful intermediate candidates, using their diagnostic results to guide the next revision. The best evaluator is maintained separately and replaced only by an improved candidate. Evolution moves to a new direction when diagnostic results provide sufficient evidence against the current one or when its search budget is exhausted.

We evaluate this framework across structured visual artifacts, images and 3D, text generation, and automated research, covering both reused preference datasets and newly collected human preferences. IPM identifies preference subsets that lead to stronger evaluator improvement than standard sampling from the same data source. Under matched search budgets, ERA continues developing directions after unsuccessful early candidates and produces more sustained improvement than the default setup. The resulting evaluators also achieve stronger agreement with held-out human preferences and improve downstream output selection and critic-guided refinement. On representative tasks, their scores further provide useful training rewards for improving generation quality.

Our contributions are as follows:

- We extend evaluator self-evolution to a broader open-ended setting, developing a unified framework that learns from human preferences and revises the full evaluation procedure across heterogeneous task domains.
- We introduce ERA, a depth-first evolution method that explicitly maintains an improvement direction across revisions, preventing an unsuccessful intermediate candidate from automatically ending development of that direction. We complement ERA with IPM for selecting informative preferences from existing labeled data or prioritizing comparisons for new human annotation.
- We provide a cross-domain evaluation of preference mining, evaluator evolution, final agreement with human preferences, and downstream utility under controlled annotation and search budgets.

2 RELATED WORK

Evaluators and data mining. G-Eval and MT-Bench study model-based judgment and human agreement (Liu et al., 2023; Zheng et al., 2023). Prometheus, Prometheus 2, and Self-Taught Evaluators improve judges through rubric-conditioned or synthetic training data (Kim et al., 2024a;b; Wang et al., 2024). AutoCalibrate refines evaluation criteria from human-labeled examples, while EvalGen helps users define criteria and select evaluation functions using human grades (Liu et al., 2024; Shankar et al., 2024). For DeepResearch reports, Lv et al. (2026) train query-specific rubric generators from human preferences and use the learned rubrics as training rewards. Our optimization target includes both the criteria and the procedure for gathering and using evaluation evidence.

RewardBench and JudgeBench assess evaluators on preference comparisons (Lambert et al., 2025; Tan et al., 2025). Benchmark² instead examines benchmark quality through ranking consistency, discriminability, and capability alignment, and studies selective test-set construction (Qian et al., 2026). Our data-mining question concerns which comparisons to select for evaluator improvement, reusing matching preferences or collecting missing ones, drawing on query-by-committee and active preference learning (Seung et al., 1992; Muldrew et al., 2024). Because our evaluation signals can measure different properties, IPM distinguishes conflicts within a dimension from trade-offs across dimensions and retains random sampling to check shared blind spots.

Optimizing language-model programs. DSPy exposes compositional pipelines to optimization (Khattab et al., 2024). ProTeGi, OPRO, TextGrad, and GEPA use model-generated feedback or proposals to improve instructions and programs (Pryzant et al., 2023; Yang et al., 2024; Yuksekgonul

et al., 2025; Agrawal et al., 2026). ADAS, AFlow, and the Darwin Gödel Machine investigate automated construction or evolution of agentic systems (Hu et al., 2025; Zhang et al., 2025; 2026). GEPA retains candidates with complementary strengths, while DGM explores an archive of agent implementations. Studying the path from experience generation through skill extraction to consumption, Huang et al. (2026) report average gains alongside negative transfer and differences between a model’s ability to extract and use skills. Their study tests whether extracted skills help a target agent; ERA examines how to continue an evaluator-improvement direction after an unsuccessful implementation, separately from accepting that candidate.

Domain quality and downstream use. SummEval and BERTScore address summarization evaluation and text similarity (Fabbri et al., 2021; Zhang et al., 2020); ImageReward, Pick-a-Pic, and GenEval capture different aspects of image preference and alignment (Xu et al., 2023; Kirstain et al., 2023; Ghosh et al., 2023). These signals are useful evidence, not interchangeable definitions of quality. ReAct, Reflexion, and Self-Refine connect observations, reflection, and iterative improvement (Yao et al., 2023; Shinn et al., 2023; Madaan et al., 2023). Human-feedback summarization, instruction tuning, and direct preference optimization connect evaluation to model learning (Stiennon et al., 2020; Ouyang et al., 2022; Rafailov et al., 2023). The reward-hacking survey of Wang et al. (2026) organizes risks of proxy optimization around objective compression, optimization amplification, and evaluator-policy co-adaptation. An evaluator’s agreement on fixed outputs therefore needs to be distinguished from its behavior under optimization, motivating separate tests of fixed-pool selection, critic-guided refinement, and reward-guided training.

3 SCOPE AND PROBLEM FORMULATION

A *domain* is a collection of tasks sharing core quality dimensions, their priorities, the overall decision rule, and the human annotation protocol. An artifact type does not define a domain: screenshot-conditioned webpage reconstruction and specification-conditioned webpage generation require different standards. A change of source distribution that preserves the standard is instead a within-domain distribution shift. Appendix A enumerates the full research scope of 22 domains in four families, without assuming that all are ready for experiments. Closed tasks whose complete quality criterion is an executable postcondition fall outside this scope.

Existing human preferences are not a domain-admission requirement. Their availability determines the missing data work, not the algorithm’s definition. An evaluation ecosystem can exist without a suitable preference dataset. We distinguish *reused preferences*, which test improvement on established benchmarks, from *new preferences*, which test applicability when matching human judgments must first be collected. One domain can support both settings.

Let x be a task with permitted context, y a candidate output, and $\mathcal{Q}_D = \{q_1, \dots, q_m\}$ the quality dimensions for domain D . A human comparison has the core form

$$h = (x, y_a, y_b, q, p, a, \nu), \quad (1)$$

where q is a dimension or overall quality, p is a preference, tie, or abstention, a records annotator agreement, and ν records provenance. Human feedback has four levels: H_0 gives overall preferences; H_1 gives dimensional preferences; H_2 identifies error type, severity, and location; H_3 optionally supplies a free-form rationale. H_1 is central to the dimension-guided design. An H_0 -only corpus remains usable, but it supplies neither human dimensional judgments nor human explanations.

The learned object is an executable evaluation program

$$E_\theta = (A, K, V, T, \Pi, G), \quad (s_\theta, \tau_\theta) = E_\theta(x, y), \quad (2)$$

comprising artifact adapters A , dimension schema and rubrics K , evaluator modules V , tools T , evidence routing Π , and fusion G . The score s_θ is accompanied by an execution trace τ_θ . The base model is fixed outside θ . Self-evolution means automated changes to this evaluation program under external human feedback, not self-definition of the human standard. At deployment, the evaluator receives one candidate output and its permitted evidence; opponents, human labels, generator identity, split labels, and mining strata are excluded. We seek agreement with unseen human preferences at an explicitly reported evaluation cost, rather than a single judge shared across all domains.

4 INFORMATIVE PREFERENCE MINING

Informative Preference Mining (IPM) lowers the sampling priority of comparisons on which existing evaluation signals consistently agree. Repeated annotations of these comparisons may add little information about evaluation errors. IPM selects comparisons from existing datasets or newly generated outputs, reusing compatible human preferences and collecting missing ones while retaining a random anchor.

4.1 COMPARABLE EVALUATION SIGNALS

Outputs are grouped by their original task input and deduplicated before comparison. For a pair $r = (x, y_a, y_b)$, signal j supplies a preference sign $u_j(r) \in \{-1, 0, +1, \perp\}$ from its own score difference and declared tie margin. Zero denotes a tie and $\perp$ missing or abstaining output; raw scores from different signals are never subtracted. Signals require executable sources with compatible inputs; provenance, repeated-run variation, and missingness are recorded separately.

We freeze a many-to-many mapping M_{jq} from signal j to dimension q using its definition and measurement procedure before reading preference labels. A mixed score may connect to several dimensions, but remains one shared signal. Signals sharing a model, training labels, or underlying metric are assigned source groups so that duplicated prompts do not become independent votes. Metrics reserved for final independent assessment remain outside mining, evaluator tools, critique, and reward.

4.2 SAMPLING PRIORITY

The ranking distinguishes three cases. *Within-dimension disagreement* occurs when signals from distinct source groups favor opposite outputs on the same criterion. *Overall-dimension disagreement* occurs when an overall signal opposes a dimensional signal. *Cross-dimensional trade-offs* occur when dimension-level preferences favor different outputs; these are not contradictory measurements of one criterion. Dimensional preferences aggregate at most one consensus sign per source group, with internally opposed groups abstaining. Writing their indicators as $I_{\text{within},q}$, $I_{\text{overall},q}$, and I_{tradeoff} , the implemented priority is

$$P(r) = 2 \sum_q I_{\text{within},q}(r) + \sum_q I_{\text{overall},q}(r) + I_{\text{tradeoff}}(r). \quad (3)$$

IPM ranks pairs by decreasing $P(r)$. This structural score does not estimate information gain or human reliability. In particular, $P(r) = 0$ can reflect missing signals or ties rather than agreement, and agreement itself need not be correct.

The random anchor precedes ranking, allowing checks for shared blind spots even among low-priority comparisons. Anchor annotations count toward the same human budget as targeted annotations. Human-reliability qualification remains separate from the implemented ranking (Appendix B). Only evolution data undergo IPM; final evaluation uses independently specified random or stratified sampling. Whether IPM produces more informative preferences is tested by subsequent evaluator improvement at equal human cost, not by the number of disagreements collected.

4.3 HUMAN FEEDBACK FOR EVOLUTION

Existing preferences can be reused when the task, readable candidate outputs, and judged quality construct match. Derived comparisons retain their ranking, rating, or error-annotation provenance. When matching preferences are absent, the task and annotation protocol are fixed before collecting blinded judgments on existing or freshly generated outputs.

The feedback table $Z(r, q)$ links artifact provenance, available human judgments, source signals, evaluator decisions, tool observations, and execution cost. Both construction settings supply this interface to ERA, while retaining their label origins and annotation protocols. Overall and dimensional preferences reveal evaluation errors for ERA to investigate, with localized diagnostics and human rationales included when available. Missing feedback levels remain absent, and model-generated explanations are not human H_1 or H_2 .

5 ERA: DEPTH-FIRST EVALUATOR EVOLUTION

ERA uses errors revealed by human feedback to search over executable evaluation programs with a fixed base model. Its central distinction is between candidate acceptance and direction continuation: an unsuccessful candidate need not replace the best evaluator for its direction to receive further work. Program inheritance separately determines which implementation that work starts from.

5.1 OBJECTIVE AND MEASUREMENT

For domain D , let $\mathcal{P}_D$ sample an input and then uniformly sample a strict human preference within its output group. With a fixed repeated-scoring protocol, the target is

$$J_D(\theta) = \Pr_{(x, y^+, y^-) \sim \mathcal{P}_D, \xi} [\bar{s}_\theta(x, y^+) > \bar{s}_\theta(x, y^-)], \quad (4)$$

where ξ includes randomness in the R executions used for each median score. Each execution is pointwise (Equation 2), without access to the opponent or preference label. The default synthesizer receives specialist findings, reasons, and scopes, but not structured scores or scales (Figure 3; Appendix E).

Search estimates agreement on a finite cohort, requesting $R \geq 3$ executions per artifact for full-evaluator measurements. For complete executions on input groups $\mathcal{S}$ with nonempty strict-preference sets P_g , define

$$\begin{aligned} \bar{s}_\theta(x, y) &= \text{median}_{r=1, \dots, R} s_\theta^{(r)}(x, y), \quad z_{gj} = \mathbb{I}[\bar{s}_\theta(x_g, y_{gj}^+) > \bar{s}_\theta(x_g, y_{gj}^-)], \\ \hat{J}_\mathcal{S}(\theta) &= \frac{1}{|\mathcal{S}|} \sum_{g \in \mathcal{S}} \frac{1}{|P_g|} \sum_{j \in P_g} z_{gj}. \end{aligned} \quad (5)$$

Input groups receive equal weight; predicted ties count as incorrect. The native development proxy uses one fixed pair per input and need not represent every within-group preference. Missing executions retain their denominators through coverage and missing-outcome bounds (Appendix F).

5.2 DIRECTIONS AND CANDIDATES

A direction d specifies a correction to the evaluation procedure, motivated by a hypothesis about how it would repair an observed error. Implementing that correction may require coordinated changes across (A, K, V, T, Π, G) . For example, correcting omission judgments can require finding missing content, assessing its importance, and making that assessment affect the score. Changing only one step may leave the error unresolved. A candidate C is one complete program implementing the direction, not an individual edit.

The agent constructs candidates from inspectable proposal data $\mathcal{D}_{\text{prop}}$, using the feedback table to connect evaluation errors to artifacts and execution traces. A disjoint sealed subset $\mathcal{D}_{\text{dev}}$ returns aggregate measurements only. The direction is the unit of continued exploration; the complete program is the unit of selection. Connected edits share one parent and one measured outcome, so a bundle’s gain is not independent evidence for each edit.

5.3 CANDIDATE ACCEPTANCE AND PROGRAM INHERITANCE

Let B_t be the best program accepted so far and W_t the working program for the active direction. A new direction starts with $W_t = B_t$; within that direction, the next candidate edits W_t using retained diagnostics. Every valid, distinct candidate receives the same development measurement protocol.

Define $\Gamma_t(C)$ as the acceptance guard: complete and comparable execution, nondecreasing coverage, and the prespecified stratum-regression check. Writing $\hat{J}_{\text{dev}}(C)$ for the recorded agreement of program C , acceptance is

$$a_t(C) = \mathbb{I}[\hat{J}_{\text{dev}}(C) > \hat{J}_{\text{dev}}(B_t)] \mathbb{I}[\Gamma_t(C)]. \quad (6)$$

A separate predicate $w_t(C)$ determines whether the candidate can be inherited as the next working program:

$$B_{t+1} = \begin{cases} C, & a_t(C) = 1, \\ B_t, & \text{otherwise,} \end{cases} \quad W_{t+1} = \begin{cases} C, & w_t(C) = 1, \\ B_t, & \text{otherwise.} \end{cases} \quad (7)$$

Set $w_t(C) = 0$ if required local or development comparisons are missing, incomplete, or incomparable. Otherwise, $w_t(C) = 1$ for accepted candidates, candidates without observed damage, or damaged candidates with fresh, previously uncredited progress and preserved protection cases. Damage means a local regression or net regression against the parent or best program on sealed development comparisons. Progress is a newly repaired root error or a predeclared exact trace-field change; neither automatically verifies the hypothesis (Appendix D). Both decisions use the pre-update state, so failed inheritance returns to B_t even if C becomes B_{t+1} .

Equation 7 preserves nondecreasing recorded development agreement for B_t while allowing W_t to pass through candidates that do not improve it. This is a selection invariant, not a guarantee of unseen agreement or statistical significance.

5.4 CONTINUING A DIRECTION

Depth-first allocation gives the active direction the next editing opportunity rather than treating candidate rejection as a reason to switch directions. The agent uses diagnostic results to distinguish an incomplete implementation from a correction that did not help despite being implemented. These diagnostics guide further edits without guaranteeing eventual improvement.

Direction continuation does not require inheriting the latest candidate. When inheritance fails, the agent can keep working on the same direction from B_t , retaining the rejected candidate’s diagnostic history. It can request diagnostics, submit another implementation, or pause when no different informative test is proposed. The controller enforces program-inheritance eligibility and resource limits, not the agent’s interpretation of failure.

Judging a direction unhelpful requires diagnostic results that contradict its hypothesis within its stated scope, after relevant implementation problems have been examined. The controller does not make this judgment automatically, and resource exhaustion leaves the direction unresolved. Adaptive reuse makes $\mathcal{D}_{\text{dev}}$ development data, not an independent test. On normal completion, Train orders eligible candidates; terminal $\mathcal{D}_{\text{val}}$ assesses the frozen shortlist without reranking or edits. The seed remains if no eligible improvement exists, and independent Test/OOD assessment follows method freeze (Algorithm 1; Appendix C).

6 EXPERIMENTS

The experiments ask whether the evaluator improves judgments and downstream outputs (C1–C3), and whether IPM and direction continuation account for those improvements under matched resources. Readiness determines inclusion from the 22-domain scope, with a separate evaluator per standard (Appendix A). Existing datasets supply candidate outputs for mining or reusable preferences; missing preferences are collected on existing or new outputs.

6.1 CORE RESULTS

C1: evaluator alignment and selection. Table 1 compares ERA with metric references, static judges, and alternative adaptation methods. Controlled learned comparisons share feedback and the initial model (Appendices G and J). For each test input, all evaluators receive the same output group, with a default design of $N = 4$. Complete human rankings support global pairwise agreement, top-region agreement involving the human top-two outputs, Best-of- N selection, and selection regret. When only ranks are available, normalized regret is $(\text{rank}(\hat{y}) - 1)/(N - 1)$ for the selected output $\hat{y}$. A pair-only dataset supports pairwise evaluation, not an N -way selection claim.

Source-disjoint ID/OOD tests preserve the evaluation standard and sample independently of mining conflicts. For a fixed evaluator, paired effects use 95% intervals over original-input groups; independent searches assess optimizer variation separately. Appendix F specifies coverage, worst strata, repeated scoring, and uncertainty.

C2: critic-guided refinement. For each unseen input, a frozen generator produces one fresh output y_0 . The seed and ERA critics inspect that same output with matched evidence and critique formats. The same generator refines each once under identical prompts, decoding, tools, and token budgets. New blinded rankings compare ERA-refined with Seed-refined and assess both against y_0 .

including major regressions. SV4 and IG1 are priorities, with TG1 testing continuity across tasks (Table 8).

C3: reward-guided training. Base, Seed-GRPO, and ERA-GRPO share an initial checkpoint, with LoRA and group relative policy optimization for the trained arms (Hu et al., 2022; Shao et al., 2024). Rewards are frozen before training. Rollouts are sampled online by each arm’s own evolving policy and are not shared. The arms match training inputs, policy-update resources, and reward-call budgets. New blinded judgments and reserved external metrics assess their fresh outputs. C3 targets IG1, SV2, and the TG4 control after reward diagnostics (Appendix I). C3 remains a design-only study here; no parameter training or training benefit is reported.

6.2 ABLATION STUDIES

Search comparisons hold feedback fixed and count all proposal, diagnostic, measurement, and validation costs. The decisive contrast is direction continuation versus history-matched proposals that always edit the best program (Figure 4a). The primary matching variable is total model-and-tool cost under frozen prices, not the number of attempts. Sampling policies are compared at equal human budgets $N_H \in \{20, 50, 100, 200\}$, measured in original-input annotation units. The endpoint is final held-out alignment under the same evolution budget, with random anchors charged to every arm (Figure 4b). The decisive sampling contrast is IPM versus random sampling with fixed feedback detail. Equal units test label efficiency; measured annotation time is needed to test human efficiency. Same-pair feedback-granularity tests and compatible component replacements complete the comparisons in Appendix J.

6.3 ANALYSIS AND STUDIES

Trajectory analysis links failed implementations, diagnoses, subsequent edits, and outcomes, including exhausted directions. Paired interventions test missing observations versus their incorrect use, while same-subset comparisons assess feedback conversion. Appendix K separates these analyses from annotation, search, and deployment costs.

7 RESULTS AND ANALYSIS

Draft status. Table 1 is unmeasured; Figure 2 uses simulated records to illustrate task-level reporting. The interpretations below are conditional, not observed findings. Downstream and budget placeholders appear in Appendices H and J.

Table 1: **C1 main comparison: unmeasured placeholders.** Pair/Top: global/top-region agreement; BoN: Best-of- N accuracy; Reg: normalized rank regret. Cells await equal-domain means and input-group 95% intervals for one fixed search replicate. Methods share cohorts per endpoint; eligible-domain counts must accompany results, with pair-only corpora entering Pair only. Dashes are pending, not zeros; shading is not a ranking. Appendix H specifies disaggregation and separate search variation.

Evaluator	ID test				OOD test			
	Pair $\uparrow$	Top $\uparrow$	BoN $\uparrow$	Reg $\downarrow$	Pair $\uparrow$	Top $\uparrow$	BoN $\uparrow$	Reg $\downarrow$
Best single metric	—	—	—	—	—	—	—	—
Metric ensemble	—	—	—	—	—	—	—	—
Static model judge	—	—	—	—	—	—	—	—
Static tool judge (seed)	—	—	—	—	—	—	—	—
Score calibration	—	—	—	—	—	—	—	—
Evaluator fine-tuning	—	—	—	—	—	—	—	—
Prompt optimization	—	—	—	—	—	—	—	—
Program search	—	—	—	—	—	—	—	—
ERA	—	—	—	—	—	—	—	—

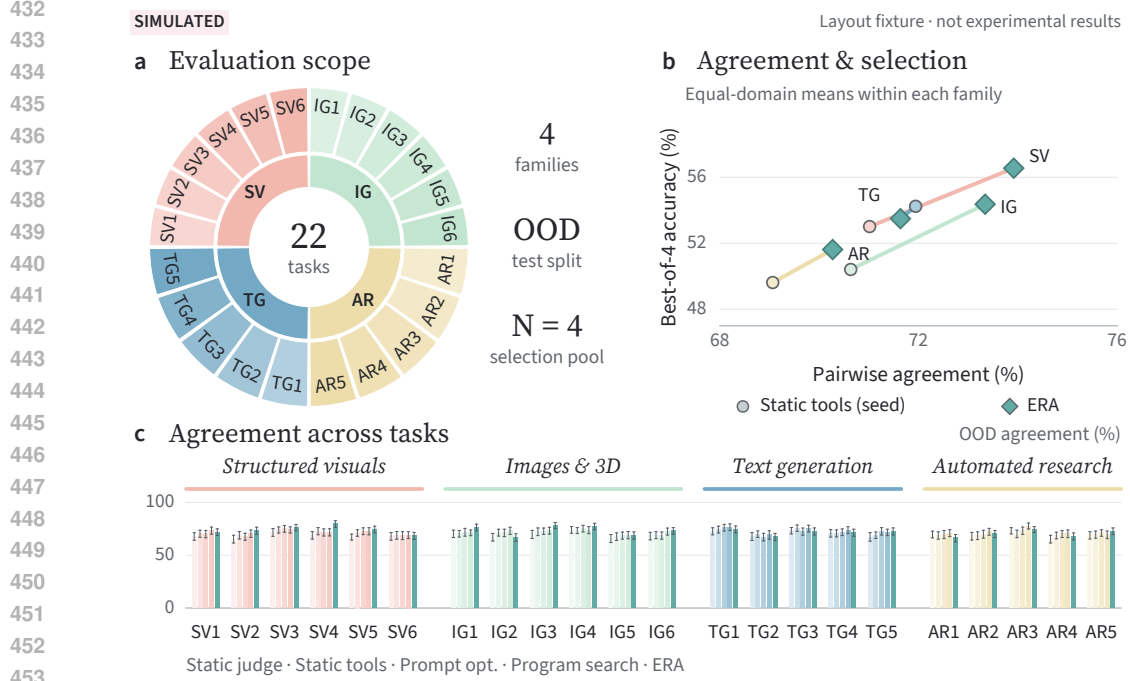


Figure 2: **C1 across tasks: simulated reporting layout, not experimental results.** Panel (a) maps 22 tasks into four families; equal family quadrants indicate taxonomy, not sample shares. Panel (b) links seed and ERA family means for pairwise agreement and Best-of-4 selection, without implying a causal relation. Panel (c) reports five methods on each task using their jointly complete OOD inputs. Whiskers are 95% input-bootstrap intervals; selection counts failed scoring as failure. Table 1 specifies the full empirical comparison. Appendix L documents denominators and provenance.

C1: agreement that improves selection. The favorable pattern would be higher ID/OOD agreement together with better Best-of- N selection and lower regret (Table 1). Figure 2b–c separates selection from task-level agreement; an agreement gain without better selection would limit the downstream claim. Cross-domain gains would support adapting the procedure to different standards, not transferring one evaluator universally.

C2: critique that improves the output. More human-preferred refinements, accompanied by fewer major regressions, would indicate that the evolved evaluator provides more useful guidance (Table 8). Changes relative to y_0 would distinguish genuine improvement from a less damaging revision.

C3: reward that improves generation. A favorable outcome would combine human preference for ERA-GRPO over Seed-GRPO with retained factuality, coverage, and validity. Higher reward alone would be insufficient; comparison with Base must also establish whether training improves generation. Conclusions would remain task- and policy-specific, with external checks reported separately (Table 9).

Continuation rather than extra search. Higher held-out agreement than restarting and history-only proposals at matched total cost would support depth-first allocation (Figure 4a). To support the proposed mechanism, trajectories must also connect a failed implementation, a diagnosed obstacle, and a later correction within the same direction.

Informative preferences rather than more labels. A favorable sampling result would reach the same held-out agreement with fewer annotation units, or higher agreement at the same units (Figure 4b). Equal-label gains that disappear at equal annotation time would not establish greater human efficiency.

8 DISCUSSION

ERA permits continuation without lowering the recorded best development score, but continuation has an opportunity cost. If history-only search beats independent proposals yet matches ERA at equal cost, retained diagnostics may explain gains without direction continuation. Development-only gains would instead suggest adaptation to repeatedly inspected feedback, not better evaluation of unseen outputs. Neither outcome supports a general advantage for depth-first allocation.

Downstream use changes which errors matter. Selection depends on ordering the best outputs; critique requires actionable diagnoses. Training shifts outputs toward whatever the reward favors, so C1 agreement or C2 refinement cannot substitute for C3 evaluation.

8.1 LIMITATIONS AND FUTURE WORK

First, consensus can hide shared errors, requiring random checks and measured human-time comparisons. Second, diagnostics do not verify a hypothesis; independent trajectory review must include failed and exhausted directions. Finally, sampling efficiency and continuation benefits remain unmeasured, requiring matched-budget runs before empirical claims.

9 CONCLUSION

Human preferences must be turned into evaluation procedures that work on new outputs. IPM targets redundant annotation; ERA separates candidate acceptance from direction continuation. Matched-budget comparisons test whether these choices improve evaluation, while separate downstream tests assess selection, refinement, and reward-guided training. The present placeholders and simulated illustrations provide no empirical evidence of these benefits.

AI USE STATEMENT

Generative AI assisted manuscript drafting, literature organization, and the preparation of editable LaTeX conceptual diagrams and plotting code. GPT Image 2 generated the photograph in Figure 1a through scene generation and a controlled edit. The quantitative displays use explicitly simulated data, generated offline for reviewing the reporting design. They are not observed cases, human annotations, model executions, or performance measurements. No empirical results or parameter-training runs were produced for this draft. Bibliographic identities were checked against primary metadata sources. Author review, responsibility for the final content, and the final disclosure of research-stage AI use remain to be completed before submission.

ETHICS STATEMENT

Human preferences can encode annotator and model biases rather than a universal definition of quality. Domain-specific criteria and uncertainty should remain visible, especially for automated research judgments. Any new annotation study must document consent, compensation, privacy protection, and the applicable institutional review determination before data collection. Artifact licenses and source permissions must be checked independently of technical accessibility. Generated code and evidence tools require restricted execution and controlled external access. Reward-guided optimization requires checks for exploitation of the evaluator and degradation outside its favored criteria. No private artifacts, identifiable annotation records, or experimental traces are included in this working manuscript. Study-specific approvals and release decisions are pending.

REPRODUCIBILITY STATEMENT

Sections 3–6 define semantic mappings, feedback, the evaluation program, evolution, and controlled evaluation. Appendices B–F specify annotation, access boundaries, the algorithm, and measurement. Appendices G–M specify adaptation controls, core reporting, planned training, ablations, analyses, simulated fixtures, and the experiment manifest. Appendix L identifies the offline generator and provenance for the numerical displays. The current manuscript contains no empirical results to reproduce. Before submission, the authors must supply a versioned, appropriately anonymized implementation and permitted data-processing and analysis artifacts with completed manifests. Their availability is not asserted here.

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A DOMAIN PANORAMA AND EVALUATION STANDARDS

The research scope is organized by evaluation standard, not by file format. Table 2 retains all 22 candidate domains. Its dimensions are a starting taxonomy to be refined from domain evaluation knowledge before selecting datasets. Each instantiated domain must additionally fix priorities, decision rules, and an annotation protocol. Inclusion in this panorama does not establish data readiness, an implemented adapter, or empirical transfer. Figure 1a gives representative artifacts; the table makes their differing evaluation criteria explicit.

Table 2: Complete domain panorama. The four families organize the scope; every row requires its own standard and readiness audit. These are study candidates, not 22 completed evaluations.

ID	Domain	Core quality considerations
Structured visual artifacts		
TD-SV1	Brief-conditioned single-slide generation	Brief coverage, hierarchy, readability, composition, and visual consistency.
TD-SV2	Screenshot-conditioned webpage reconstruction	Reference content, spatial arrangement, typography, and visual fidelity.
TD-SV3	Data-to-visualization generation	Data fidelity, valid encodings, readability, and communicative usefulness.
TD-SV4	Specification-conditioned webpage generation	Requirement satisfaction, functional behavior, usability, and visual organization.
TD-SV5	Source-grounded slide-deck generation	Source faithfulness, narrative, coverage, slide readability, and deck consistency.
TD-SV6	Topic-conditioned slide-deck generation	Relevance, factual support, narrative completeness, audience fit, and presentation quality.
Image generation and understanding		
TD-IG1	Detailed image-caption generation	Grounding, coverage, relation accuracy, specificity, and absence of hallucination.
TD-IG2	Text-to-image generation	Prompt alignment, object and relation accuracy, visual coherence, and preference.
TD-IG3	Single-source instruction-based image editing	Instruction fulfillment, preservation of unrequested content, and image quality.
TD-IG4	LR-conditioned image super-resolution	Reference fidelity when available, perceptual detail, artifact avoidance, and plausible texture.
TD-IG5	Text-to-3D object generation	Prompt correspondence, geometry, appearance, and multi-view consistency.
TD-IG6	Image-to-3D object generation	Source-view fidelity, plausible unseen geometry, appearance, and multi-view consistency.
Text generation		
TD-TG1	Source-grounded document summarization	Faithfulness, relevance, coverage, coherence, and concision.
TD-TG2	RAG-supported long-form answer generation	Answer correctness, evidence grounding, attribution, completeness, and organization.
TD-TG3	Machine translation	Meaning preservation, omissions and additions, fluency, terminology, and style.
TD-TG4	Instruction-constraint-following response generation	Constraint satisfaction, substantive correctness, relevance, and response quality.
TD-TG5	Open-ended dialogue-response generation	Contextual appropriateness, helpfulness, consistency, safety, and conversational quality.
Automated research		
TD-AR1	Research-idea generation	Novelty relative to evidence, significance, feasibility, and testability.
TD-AR2	ML/NLP ablation-plan generation	Hypothesis identification, controlled contrasts, confound management, and feasibility.
TD-AR3	Table-to-insight report generation	Numerical faithfulness, appropriate inference, uncertainty, and relevance.

Table 2 continued

ID	Domain	Core quality considerations
TD-AR4	Related-work generation	Coverage, citation correctness, faithful comparison, and synthesis.
TD-AR5	Peer-review critique generation	Technical accuracy, evidence, significance assessment, fairness, and actionable criticism.

Representative resources include PPTAgent and PresentBench for presentations, Design2Code for screenshot-conditioned webpages, SummEval for summarization, and image-preference and alignment resources (Zheng et al., 2025; Chen et al., 2026; Si et al., 2025; Fabbri et al., 2021; Kirstain et al., 2023; Xu et al., 2023; Ghosh et al., 2023). Their outputs, references, metrics, and annotations have different evidential roles; a resource is not automatically a source of the required human preference. A benchmark collection spanning several standards remains a dataset collection, not an additional domain. Research execution with a complete executable postcondition is excluded, while ideation, study design, interpretation, writing, and critique remain open-ended evaluation problems.

Dataset roles. Dev supports data mining and evaluator development, ID tests unseen inputs from the development source distribution, and OOD changes source distribution while preserving the standard. External datasets provide separately locked assessment only and do not determine semantic mappings or data mining. Harder examples from the same source are a difficulty stratum, not a new OOD domain. A domain selected for confirmatory execution must pass the audit below; selection does not remove unselected domains from the research scope.

B FEEDBACK PROVENANCE, ANNOTATION, AND READINESS

The two construction settings in Table 3 share the same evaluator-evolution problem. They differ in the work needed to obtain matched human feedback. A domain may use both settings. Direct, derived, and materials-only provenance describe what a resource supplies; they are not additional domain classes or alternative algorithms.

Inputs, references, editable artifacts, and automatic metrics without matched human judgments are materials only. They may support output construction or independent assessment, but do not supply human feedback. When H_0 is absent, we preserve the original task and evaluation standard, separate source groups, recover readable outputs or generate them with frozen generators, and collect the required feedback. Preparation can precede annotation; final human-alignment claims cannot precede matched human assessment.

Annotation units and levels. One annotation unit is an original input and its anonymized output group, including permitted references. Pairs derived from that group remain dependent. The core tuple in Equation 1 is retained together with dataset and license, source grouping, generator and prompt version, collection time, annotator pool, agreement, and conversion provenance. Generator identity is recorded for audit but hidden from annotators and the deployed scorer.

The planned evolution protocol collects H_0 and core H_1 , with H_2 on conflict and failure subsets. H_3 is optional. Two independent annotators with adjudication may support development; final ID/OOD assessment targets three independent annotators and blinded complete rankings. Core dimensions and the tie policy are fixed in advance. A model or an author decision is not substituted for a missing independent final annotation. An H_0 -only setting must be reported as such; the framework’s richer interface does not establish that a run actually received H_1 or H_2 .

Agreement, abstention, and reannotation. New labels are not assumed to be more reliable than reused labels. Report the agreement statistic, its valid-response denominator, and the number of raters per group. Abstention uses all requested ratings as its denominator, including those excluded from valid-pair agreement. Group outcomes distinguish decisive preference, tie, no consensus, and insufficient votes. The simulated audit in Figure 8a–c illustrates these separate quantities. Individual disagreements are retained rather than filtered to match an old winner.

Table 3: Two routes to human feedback. Report each source’s protocol, agreement, abstention, and available human feedback separately. An adapter changes representation, not the quality construct or the provenance of a judgment.

Setting	Label origin and protocol	Human feedback
Reused preferences	Original human pairs/rankings on readable same-input outputs; retain the source protocol and document any conversion from rankings, scores, or errors	H_0 only for overall judgments; H_1/H_2 only when actually annotated. Unknown source agreement or abstention remains unknown
New preferences	Freeze the task, visible evidence, dimensions, ties, and abstention before collecting independent blind judgments on existing or freshly generated outputs	Collect the levels required by the study; preserve individual votes and unresolved disagreements. Fresh C2/C3 outputs require new endpoint judgments

Annotators do not see model scores, method identity, or earlier winners. A changed rubric or visible evidence defines a changed protocol; resulting differences do not by themselves establish errors in the older labels.

Label-blind IPM. Freeze domain knowledge, signal definitions, semantic mappings, and the calibration recipe before accessing preference labels. Select a random anchor, reveal only its authorized training judgments to estimate reliability, then freeze the targeted sampling policy before opening labels in the remaining pool. Charge anchor annotations to every strategy’s human budget. An existing labeled dataset must simulate the same access order. The final cohort is sampled independently of conflicts and never contributes to reliability calibration.

Implemented conflict ranking. IPM computes structural disagreement without reading the pair’s human preference. Let $I_{\text{within},q}(r)$ indicate opposing signals from distinct source groups on dimension q , $I_{\text{overall},q}(r)$ indicate opposition between an overall and a dimensional signal, and $I_{\text{tradeoff}}(r)$ indicate opposed dimension-level preferences. Equation 3 ranks these conflicts, assigning twice the weight to within-dimension disagreement. Pairs are ranked by decreasing $P(r)$, with stable pair identifiers breaking ties. The coefficients define a structural ranking, not estimated signal reliability or a probability of evaluation error. Overall–dimension comparisons retain whether their signals share a source group; they are not automatically independent evidence. Each source group contributes at most one preference sign to a dimension’s aggregate preference, with opposing within-group signs yielding abstention. Missing outputs and small-margin ties remain distinct from opposing preferences.

For repeated model signals, the pairwise signal margin is the median of same-repeat score differences between the two artifacts; variation and sign changes are retained. This differs from Equation 5, which first takes each artifact’s median final score and then compares them. A missing required signal repeat makes that evaluation signal unavailable for mining. Deterministic metrics use declared ranges and orientations, not ranges fitted to the output pool or human labels. The random anchor is selected by original-input hashes before scoring, preserving output groups. Label-blind execution does not establish label-blind upstream cohort selection; historical data must also document how the output pool was chosen.

Human qualification and implementation boundary. The implemented miner supplies the structural queue and its source-dependence records. The full IPM protocol additionally requires human-annotated anchors to qualify signals before targeted annotation; cross-domain human-reliability calibration and the equal-budget advantage remain unverified. Model-annotated pilot anchors measure agreement with that model, not human reliability. The annotation fraction, qualification thresholds, and targeted policy must be frozen for the human study, rather than inherited silently from a pilot. The feedback table Z is a logical join over feedback, artifact, signal, and execution records; absent human dimensions or rationales remain absent.

Source and tool status. Registry entries distinguish executable, planned, and blocked sources (Table 4). A publication or code link without compatible inputs, weights, or dependencies is not an executed signal. Available tools begin as candidates; a route must explicitly activate them. Source

Table 4: Readiness criteria before a domain enters confirmatory evaluation. These are audit requirements, not completed coverage scores.

Audit object	Required record and exclusion condition
Evaluation standard	Open-endedness, dimensions and hierarchy, priorities, decision rule, and human protocol. Exclude tasks fully decided by an executable postcondition.
Knowledge provenance	Benchmarks, expert guidance, rubrics, and methods with versions, dates, input assumptions, known biases, and availability.
Source coverage	Dimension $\times$ source-type matrix: deterministic, learned metric, LLM/VLM judge, external analysis tool. Each core dimension needs executable evidence.
Source independence	For conflict mining, ordinarily two mechanisms per dimension. Record shared models, training labels, metrics, and mixed signals; multiple names are not independent votes.
Executable artifacts	Recoverable input/output groups, readable artifacts, references, permissions, and source fingerprints. Missing assets do not become scorer disagreement.
Conflict pool	Three semantic strata, availability, margins, repeat variance, anchor reliability, dependence caps, and uncovered strata. Noise alone is not a gap.
Feedback fidelity	Direct versus derived provenance, agreement, mapping to $H_0/H_1/H_2$, adapter checks on a common subset, and additional annotation needs.
Source isolation	Original-input deduplication, Dev/ID/OOD/External roles, contamination review, and a pre-search External Metric Reserve.

Table 5: Information available to evolution and assessment. Every invocation of the deployed scorer remains pointwise and label-blind, including on Train.

Partition	Information released	Permitted decision
Proposal Train	Authorized artifacts, feedback, and traces	Mechanism hypotheses and candidate evolution
Sealed Train	Prespecified aggregates only	Candidate acceptance, program inheritance, and direction continuation
Terminal Val	Frozen-shortlist measurements	Terminal assessment; no reranking or further edits
Test/OOD	Locked until method and analysis freeze	Independent final reporting
Metric reserve	No access during evolution	Final assessment only; no tools, critique, reward, or selection

independence is assessed at the signal level, including model, training-label, and metric dependencies. Model-only comparisons hold rubric, evidence, context, and reasoning budget fixed. Replacing both the scoring model and its evidence-collection agent instead compares full systems. Deterministic or fixed learned metrics are not duplicated for every judge model.

C DATA ACCESS AND INDEPENDENT ASSESSMENT

Table 5 separates proposal evidence, adaptive development measurements, and terminal assessment. The scorer always receives the same pointwise input contract, regardless of the cohort in which it is invoked. Only the authorized diagnostic process sees human labels and pair identities.

Training and validation. Proposal-Train evidence may inform evolution directions, candidates, and allocation. Sealed Train reveals prespecified aggregate agreement, coverage, and stratum guards without examples, labels, or per-case errors. Aggregates remain adaptive development evidence, not a final performance estimate. The terminal shortlist and its order are fixed from training evidence before Val access, and every shortlisted outcome is reported. Repeated Val calls can assess stability but do not create an independent test cohort. Interrupted searches do not obtain a new terminal assessment merely because they stopped.

External metrics. The External Metric Reserve is frozen before search. Reserved metrics and their equivalent wrappers are excluded from the executable tool library, router, diagnostician, critique, reward, and checkpoint-selection criteria. If an endpoint was used for optimization, it loses

independent-assessment status and must be replaced or explicitly reported as an optimization metric. This restriction does not forbid using other existing metrics as tools.

D ERA ALGORITHM AND ATTRIBUTION CONTRACT

Figure 3 separates the editable observation and judgment mechanisms. Algorithm 1 specifies their evolution and selection.

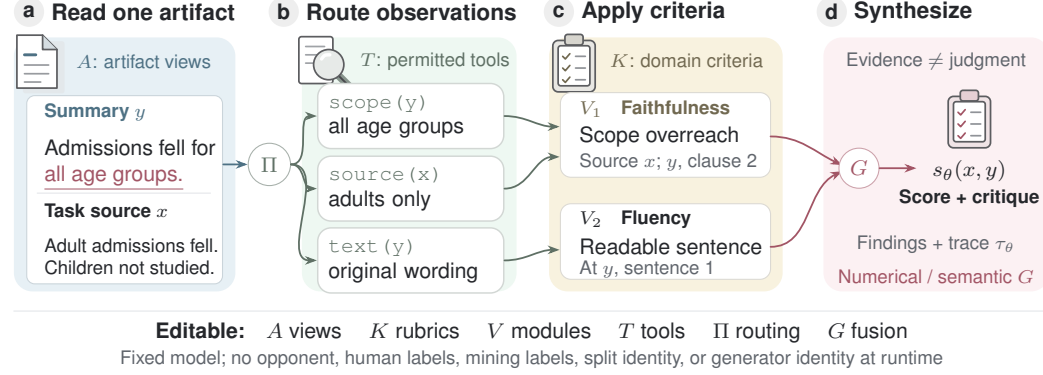


Figure 3: The six-component evaluation program. (a) Read one output and permitted task evidence. (b) Route tool observations. (c) Apply dimensional criteria. (d) Fuse sourced findings into a score and critique. The lower rail identifies the editable components. Human feedback guides evaluator evolution, never the deployed scorer. Artifact evidence and judgment remain separate; the summary and findings are a constructed example.

Algorithm 1 ERA: separate candidate acceptance from direction continuation

Require: Seed E_0 , proposal feedback table Z_{prop} , sealed $\mathcal{D}_{\text{dev}}$, fixed search and execution limits

- 1: Measure E_0 ; initialize best program $B \leftarrow E_0$, working program $W \leftarrow E_0$, and history
 - 2: **while** search resources remain **do**
 - 3: Start a direction from B , or continue the active direction from W
 - 4: Inspect proposal-Train artifacts and history; request diagnostics as needed
 - 5: Construct complete candidate C with its intent and inspected Train references
 - 6: **if** C is valid and distinct **then**
 - 7: Measure C on $\mathcal{D}_{\text{dev}}$; retain local diagnostic outcomes
 - 8: Save pre-update best $B_{\text{old}} \leftarrow B$; assess acceptance and inheritance against this state
 - 9: **if** the candidate-acceptance conditions hold **then**
 - 10: $B \leftarrow C$
 - 11: **end if**
 - 12: $W \leftarrow C$ if measured inheritance is eligible; otherwise $W \leftarrow B_{\text{old}}$
 - 13: **end if**
 - 14: Retain changes, outcomes, failed attempts, and resource use in history
 - 15: Give the active direction the next opportunity to request diagnostics or submit an edit
 - 16: Pause on the agent’s explicit request or an enforced resource/execution limit
 - 17: **end while**
 - 18: On normal completion, freeze and order eligible candidates using Train only
 - 19: Assess a nonempty shortlist on terminal Val without reranking or further edits
 - 20: Retain the Train-selected evaluator, or the seed if no eligible improvement exists
-

Investigation and candidate construction. The evolution agent reads original proposal-Train artifacts, installed component instructions, execution traces, and relevant history through a tool-driven editing session. It can request additional authorized proposal-Train measurements or fixed-input component probes. The submission contains actual edits, a short intent, and references to inspected training cases. Predicted orderings, contrast cases, and intermediate checks are optional declarations, not controller-inferred scientific predictions. The controller records the exact parent, installed program, and measured outcomes. Local diagnostic success is not an admission requirement for the common development measurement.

An elementary edit changes one component; a connected candidate can assemble a tool, its route, and its consumer against a fixed parent. Only the complete candidate is executed and measured, not its partially assembled versions. Changed tools must be connected to valid routes, although valid routing alone does not establish correct use of their outputs. Interface validity, fixed-model constraints, and execution limits remain mandatory; edit-character counts are recorded as guidance rather than a hard acceptance or stopping rule. Unchanged programs and verified historical replays are distinguished from fresh trials, and failed calls remain in the cost record.

Diagnostic results. Fixed-input probes compare supported specialist or synthesis variants while holding their upstream inputs fixed. Such probes bind their observations to the exact tested candidate and inputs; changing a tool, route, or topology requires the corresponding complete execution. An optional exact intermediate check records whether a declared field changed to its expected value on at least one complete repeat. All repeat-level matches, contradictions, and unknowns remain visible. A semantic expectation is not automatically verified, and a changed recorded field is not proof of a correct causal explanation or improved overall judgment.

Candidate acceptance and program inheritance. Candidate acceptance requires complete, comparable execution, a strict development-agreement gain over the best evaluator, nondecreasing coverage, and the prespecified worst-stratum guard. These are development selection rules rather than statistical significance claims. Program inheritance is a separate decision. Inheritance first requires known, complete, comparable local and development outcomes, including comparisons against both the parent and best program. Subject to that requirement, accepted candidates and candidates with no observed damage can be inherited. For this decision, damage includes a local ordering regression, deterioration on a declared protection case, or more regressed than corrected sealed-Train comparisons against either the parent or the best evaluator. An unaccepted candidate with damage is eligible only when fresh measurement supplies new progress and the declared protection cases do not deteriorate.

Qualifying progress is either a first-time repair of an error present at the direction’s root, still wrong in the parent, or a newly observed, predeclared exact intermediate-field change. Previously credited repairs and checks are retained across attempts, so breaking and repairing an already corrected case cannot renew this allowance. A missing protected case is unknown, not preserved. When inheritance fails, the next parent returns to the pre-update best program, even if the candidate has just been accepted as the new best. The candidate and its diagnostics remain available. Keeping an intermediate candidate never changes the recorded best score.

Direction continuation and termination. In the native editing session, the agent can request diagnostics, submit a candidate, repair an invalid submission, or explicitly pause with its unresolved question. The controller enforces parent eligibility, validity, and resource limits; a failed inheritance check returns the editable program to the pre-update best without erasing the direction’s history. The next editing opportunity remains within the direction until a pause or an execution/resource limit ends that allocation. The controller does not automatically judge whether a direction is unhelpful, and the native workflow does not require a separate model reviewer to certify direction continuation. That judgment requires diagnostic results contradicting the hypothesis after checking for missing observations, unused tool output, or incorrect interpretation. One unsuccessful implementation, a nonsignificant difference, or budget exhaustion alone is insufficient. These requirements govern interpretation of diagnostic results, not an automatic hypothesis test.

Terminal assessment and attribution. The terminal shortlist includes eligible, previously accepted, complete programs with positive development gain over the seed and is ordered using Train evidence alone. Its membership and order are frozen before Val access. Terminal Val does not rerank the shortlist or guide another proposal. With no eligible improvement, the seed is retained without an additional Val query. Fresh system confirmation and independent Test/OOD assessment remain separate from adaptive development.

A connected candidate receives one system-level outcome, not independent causal credit for each component. Component attribution requires paired interventions with the remaining interfaces fixed, such as tool on/off or evolved versus seed synthesis. If removing a dependency makes the program invalid, the comparison must report that limitation rather than substitute an unrelated whole-program

control. Complete prompt templates, interaction traces, candidate changes, and execution settings belong in the reproducibility record rather than the main algorithm.

E SEMANTIC SYNTHESIS AND ANCHORED SCORING

The default fusion procedure forms an overall judgment from specialist findings; a declared numerical rule is an explicit alternative. Each specialist produces findings with identifiers, supporting evidence, dimensions, and uncertainty, as well as its own rating and reason. The full specialist trace remains available for diagnosis, but the synthesizer’s default input omits the structured specialist score, rating, and score scale. It retains the findings, reasons, and scope of each specialist and applies its own overall scale. This input separation does not remove value judgments expressed in the reasons.

The synthesizer groups findings by their consequences for quality, including strengths, weaknesses, mixed effects, and uncertainty. Every finding reference must occur in exactly one group. This accounting rule makes omitted or multiply referenced findings detectable; it does not guarantee correct semantic grouping. Duplicate descriptions should not become independent penalties, and favorable style should not cancel a factual error when the domain standard prioritizes faithfulness.

Let $a_k < a_{k+1}$ be adjacent overall anchors and $\lambda \in [0, 1]$ locate quality between them. The resulting scalar is

$$s = (1 - \lambda)a_k + \lambda a_{k+1}. \quad (8)$$

The anchors have domain-specific verbal meanings; specialist scores with different meanings are not averaged. Reference completeness, semantic correctness, and calibration require separate checks. A synthetic arithmetic example uses $a_k = 50$, $a_{k+1} = 75$, and $\lambda = 0.4$, giving $s = 60$. These numbers illustrate interpolation, not a universal scale or an evaluator result.

F MEASUREMENT AND STATISTICAL ANALYSIS

Completeness and denominators. Use the original-input weights $w_{gj} = 1/(|S||P_g|)$ from Equation 5, fixed before execution. Let $c_{gj} = 1$ when both artifacts have all R valid complete scores, $M = \sum_{g,j} w_{gj}c_{gj}$ be covered weight, and $K = \sum_{g,j} w_{gj}c_{gj}z_{gj}$ be correct weight. Report conditional agreement K/M when $M > 0$, weighted coverage M , raw completion counts, and the missing-outcome interval

$$[K, K + 1 - M]. \quad (9)$$

For n equally weighted pairs with m complete and k correct, this reduces to $[k/n, (k + n - m)/n]$. The interval bounds full-cohort agreement under arbitrary missing outcomes; it is not a statistical confidence interval. A missing repeat is not replaced by the median of surviving calls. Completeness includes selected routes and tools, not only a final score. A valid observation that an artifact fails a quality check differs from an execution error.

Ranking and selection endpoints. For the default $N = 4$ design, human complete rankings support all strict pairs. Global agreement averages within input groups before aggregation across inputs. Top-region agreement uses prespecified strict comparisons involving at least one human top-two output. Best-of- N measures selection of a human top-ranked output; normalized rank regret is defined in Section 6.1. Human ties, evaluator ties, and incomplete rankings are governed by a frozen policy and reported separately. Pair-only data do not supply missing ranks or cardinal utility. Core H_1 agreement is reported by dimension before any macro average. Groups without a strict human preference are ineligible for strict agreement and retain a separate count; they are not assigned artificial pair labels. Repeated appearances of an artifact use its frozen scores, so pairs sharing an input or artifact are not independent observations.

Randomness and repeated use. Freeze $R \geq 3$, score precision, retry policy, artifact reuse, and cache identity. The target in Equation 4 averages over this fixed execution protocol as well as the input distribution; a finite set of recorded executions is one realization, not the population median. Retries are outcome-blind and uniformly applied. Repeated scores are not independent task observations. Paired uncertainty uses original-input/source group resampling, preserving output groups

and comparisons; the planned interval level is 95%. Report independent search seeds separately from scoring repeats, and distinguish within-run execution variation from between-search variation. Retain failed, rejected, incomplete, and interrupted attempts in the authorized analysis, without favorable resampling or denominator changes.

C2/C3 human outcomes. Report wins, losses, ties, invalid outputs, and major regressions per input group. A prespecified tie-aware preference probability may assign half credit to a tie, with its denominator and missingness bounds stated. Report C2 changes against the shared y_0 in addition to the critic contrast, and C3 changes against Base in addition to the reward contrast. Intervals and costs accompany effect sizes. Multiple domains, dimensions, and ablations require declared primary contrasts and a prespecified multiplicity policy. A search-selected development maximum is not substituted for any final endpoint.

G PROGRAM EVOLUTION AND PARAMETER ADAPTATION

Parameter adaptation is a competing approach to learning human evaluation standards, not an assumed failure case. Table 6 distinguishes score calibration, evaluator fine-tuning, and program evolution by their editable objects. Fine-tuning can learn criteria and tool-use behavior when its training interface permits them. The fixed-structure control below is a declared experimental restriction, not a claim about the limits of fine-tuning in general.

Table 6: Adaptation choices for the planned C1 comparison. Rows specify what changes, not a ranking of effectiveness. All use the same permitted human feedback; observed evidence and total adaptation cost must also be reported.

Approach	Adapted object	What the comparison tests
Score calibration	Combination weights, scale maps, or thresholds over frozen seed signals; model and evidence path fixed	Whether the preference mismatch can be addressed by a different numerical decision rule.
Evaluator fine-tuning	Judge weights from the common base checkpoint; seed adapters, tools, routes, and output interface fixed	Whether learning a new mapping from the same observations to judgments is sufficient.
ERA program evolution	Artifact adapters, criteria, evaluator modules, tools, routing, and fusion; base-model weights fixed	Whether changing the evaluation procedure improves alignment beyond parameter adaptation.

Common human feedback and selection. All arms use the same proposal-Train inputs, artifacts, and available human feedback levels. A fine-tuned evaluator learns pointwise scores from the permitted comparisons, with dimensional feedback when available; missing human explanations are not replaced by extra teacher labels in this contrast. Scoring remains pointwise and label-blind. Sealed Train supplies aggregate measurements only, never per-example training targets or gradients. Tuning configurations and checkpoints follow the same terminal-Val and locked-Test/OOD boundary as program candidates. Human units, annotation time, unsuccessful trials, optimization compute, tool calls, and selection costs are all counted. Adaptation and deployment costs are reported separately, with matched-budget comparisons wherever feasible and unmatched resources explicitly identified.

Separating observation from interpretation. The end-to-end comparison permits ERA to change its observation path. It cannot by itself attribute a difference to better interpretation of identical evidence. A shared-observation control therefore freezes (A, T, Π) and supplies the same artifact evidence to fine-tuning and ERA restricted to (K, V, G) . This control, together with the component comparisons in Table 11, tests whether changed observations account for any advantage. Neither lower annotation cost nor superiority to fine-tuning is assumed. A combined weight-and-program arm would be needed to test complementarity.

Table 7: Required empirical outputs beyond the simulated reporting examples. Every result requires authorized frozen data and a traceable analysis; neither conceptual illustrations nor simulated displays supply such evidence.

Output	Required summary	Interpretation limit
C1 alignment	Global/top-region agreement, per-dimension H_1 , worst stratum, ID/OOD, coverage, repeated-score variation	Pairwise improvement does not establish selection or generation gains.
Feedback quality	Label origin, source protocol, annotator pool, agreement, abstention, unresolved judgments, and annotation time per setting	Newly collected labels are not automatically more reliable.
C1 selection	Best-of- N , normalized rank regret, complete human rankings, grouped intervals	Rank regret is not cardinal utility; pair-only corpora are insufficient.
C2 refinement	New blinded contrast, changes against y_0 , major regressions, dimensional quality, external metrics	Fresh shared initial outputs and matched refinement are necessary.
C3 training	New blinded reward contrast, Base sanity check, external metrics, validity, reward-hacking diagnostics	Design-only until separately authorized training and final evaluation occur.
Efficiency	Human units and time, search calls/tokens/tool use, runtime, deployment and training costs	More compute or annotation detail is not a free method gain.
Trajectory	Directions, implementation attempts, continuation/pause, unresolved stops, resource allocation	A schematic tree is not an optimization trace or evidence of efficiency.
Mechanism	Paired component effects, interactions, remaining gap to human agreement, scoped counterexamples	Assign limitation regimes only after evidence; do not generalize from one stratum.
Adapter fidelity	Direct/derived label agreement and downstream differences on the same subset	Annotation conversion cannot change the construct being evaluated.

Inspectable does not mean faithful. An explicit rubric and evidence trace make the stated basis of a judgment available for inspection. They do not establish that the judgment follows that basis or matches the human standard. Mechanism assessment must check cited observations and test predicted decision changes under controlled evidence or component interventions. These tests are distinct from preference agreement.

The tuning comparison concerns the *evaluator*. C3 instead trains a *generator* against frozen evaluator rewards. Neither experiment has been executed in this draft. The simulated C1 displays illustrate a subset of reporting conditions and contain no fine-tuning or calibration outcomes.

H DETAILED CORE RESULTS AND EXTERNAL ENDPOINTS

Empirical plots for the outcomes in Table 7 require frozen reporting cohorts and analysis. Suitable outputs include human-budget curves, per-domain component effects, annotated search trajectories, and C2/C3 regression breakdowns. The numerical displays in Appendix L are reporting fixtures, not provisional substitutes for those measurements. Table 8 records the empirical status of the downstream studies independently of the simulated C2 illustration.

C1 disaggregation. Each populated main-table result will be accompanied by domain, feedback origin, split, original-input count, output count, and completion rate. Within the same registered cohort, report overall and dimensional agreement, top-region agreement, selection accuracy, rank regret, worst-stratum agreement, and repeated-score variation. Report equal-domain summaries separately for reused and new preferences, with their paired ERA-minus-baseline effects. A main-table summary uses equal-domain weights and a common registered domain set across methods for each endpoint. Report the eligible-domain count for each endpoint; pair-only domains cannot silently change the domain set used for selection metrics. Missing domains remain visible rather

than being removed according to performance. Pair-only corpora contribute only to the pairwise endpoint. Per-domain plots replace the simulated fixtures only after the underlying runs and grouped intervals are verified. Table 1 is conditional on a fixed, preregistered search replicate rather than the best replicate selected after evaluation. Its paired intervals resample original-input/source groups within domains with the evaluators fixed. Report independent-search means, dispersion, and run counts alongside this fixed-evaluator comparison; scoring repeats do not increase the independent-search count.

Table 8: **Downstream human outcomes: unmeasured placeholders.** C2 compares ERA-refined with Seed-refined; C3 compares ERA-GRPO with Seed-GRPO. Preference cells await tie-aware estimates, input-group 95% intervals, and counts. Each ΔReg cell is the ERA-minus-seed difference in major-regression rate against y_0 (C2) or Base (C3), with ID and OOD separate. Appendix H specifies per-arm rates and completion. Dashes await new blinded judgments; C3 is design-only.

Study	Domain / role	ID pref. $\uparrow$	OOD pref. $\uparrow$	ID ΔReg $\downarrow$	OOD ΔReg $\downarrow$
C2	SV4: specification-to-web	—	—	—	—
	IG1: detailed image captions	—	—	—	—
	TG1: summarization	—	—	—	—
C3	IG1: primary candidate	—	—	—	—
	SV2: second candidate	—	—	—	—
	TG4: verifiable control	—	—	—	—

C2 output comparisons. For each input, retain the fresh y_0 , Seed-refined output, and ERA-refined output as one blinded ranking group. Report the ERA-versus-seed preference with ties and abstentions separate, each arm’s improvement and regression relative to y_0 , major regressions, and core H_1 changes. For Table 8, each regression contrast subtracts the seed arm’s major-regression rate from the ERA arm’s rate within one domain and split. The C3 contrast uses the same definition relative to Base, rather than relative to a C2 starting output. Per-arm rates, paired uncertainty, and missing-outcome bounds accompany both contrasts. The generator, critique interface, observation access, decoding, and revision budget remain identical across arms. Benchmark outputs can support pilots, but do not replace fresh generation in the confirmatory comparison. IG2 and AR1 enter only after new human or expert assessment is ready; their absence is not evidence of failure.

Table 9: External endpoints for C1–C3, separate from human preference. All outcome cells are unmeasured. Specific metric versions and applicability must be frozen before final evaluation; no average is taken across different quality constructs. Reserved checks never become mining signals, evaluator tools, critics, or rewards.

Domain	Targeted attribute	Checks and protected attributes	Outcome
SV4	Specification satisfaction	Functional checks; visual usability and no lost required content	—
IG1	Caption grounding and detail	Factuality, hallucination, and content coverage; length alone is insufficient	—
TG1	Summary faithfulness	Source support and key-content coverage; protect against omission	—
SV2	Reconstruction fidelity	Render comparison and executability; visual similarity alone is insufficient	—
TG4	Verifiable quality control	Applicable reference or executable checks; do not generalize to every open-ended task	—
Other ready domains	Domain-specific standard	Report the applicable reserved check, or explicitly record no suitable metric	—

Table 9 complements human outcomes rather than replacing them. When the two disagree, inspect the same outputs, dimensional judgments, and metric scope before attributing the difference to reward exploitation or metric failure. Distinguish an unrun experiment, an inapplicable metric, and a completed experiment without confirmed improvement.

I PLANNED C3 TRAINING PROTOCOL

C3 is specified to test reward usefulness independently of evaluator agreement and critic usefulness. It is not an executed experiment, and no parameter training is part of this manuscript preparation.

Representative tasks and pre-training diagnostics. IG1 is the primary training candidate, SV2 a possible second task, and TG4 a verifiable control; AR1 is not a priority for the first training study. Before training, score a common diagnostic output pool from the actual target policy with both frozen evaluators. Check top-region preferences against new human judgments, repeated reward variance, ties, invalid scores, and sensitivity to length, omission, and format. Only this preflight diagnostic pool is shared: after updates begin, each policy generates its own online rollouts. The reward version, score mapping, tie and invalid-output policy, and Train/Val/Test separation must be frozen before policy updates. A domain proceeds only after prespecified stability and exploitation checks pass.

Table 10: Controls for the planned LoRA + GRPO comparison. Numerical hyperparameters, pilot thresholds, and final sample sizes remain to be frozen in the experiment manifest.

Control	Contract
Arms	Base is untrained; Seed-GRPO and ERA-GRPO start from the identical base checkpoint.
Training data	Same unlabeled training inputs and order; source-level deduplication and contamination audit.
Policy updates	Same LoRA structure, optimizer, learning rate, update count, group size, KL rule, checkpoint cadence, and policy-training compute.
Online samples	Each updated policy samples its own rollout groups. Do not share trajectories once policies diverge.
Reward version	Freeze program, prompt, tools, routing, aggregation, normalization, ties, abstention, and invalid-output policy before training.
Reward resources	Equal reward-call counts and token caps; report actual inference time, tool use, and monetary or compute cost separately.
Training validation	Input-disjoint development validation selects checkpoints and checks reward exploitation; final metrics and final human labels remain locked.
Pilot gate	Prespecified acceptable validity, reward variance, KL, length, stability, held-out quality, and reward-hacking diagnostics before expansion.
Final generation	Freeze checkpoints and decoding, then generate new Base, Seed-GRPO, and ERA-GRPO outputs on the same unseen ID/OOD inputs.
Final judgment	New blinded H_0 and core H_1 , reserved external metrics, invalid/major-regression rates, length and format shifts, and total cost.

Under the controls in Table 10, the frozen reward produces scalar scores within each rollout group using the same normalization and advantage rule. Zero-variance groups, invalid artifacts, and execution failures require a common prespecified policy. Deterministic checks may enforce format or executability, but do not define the entire reward for open-ended quality.

The pilot uses development validation only. If its prespecified stability or reward-hacking gate fails, expansion stops for that domain and the failure is retained. Success elsewhere is not a substitute. Final sample sizes are determined from pilot variance and a declared minimum detectable effect before test generation. Test inputs and judgments do not participate in training, checkpoint selection, reward design, or failure-driven edits.

Any deliberate sharing of locked input pools across C1/C2/C3 must be prespecified with an access audit; their newly generated outputs and judgments are never treated as reusable old labels.

J FULL ABLATION MATRIX

C1 comparator contract. The rows in Table 1 are controlled comparison specifications, not claims that every baseline configuration has been executed. The best single metric is selected per domain using Train only from a prespecified applicable metric set, with score orientation fixed

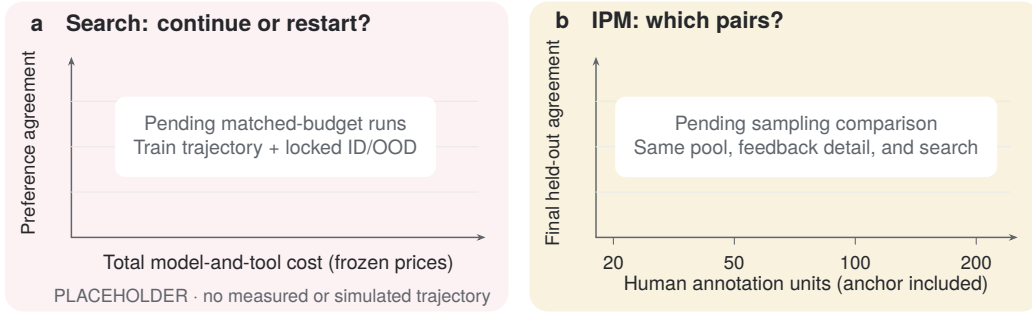


Figure 4: **Matched-budget comparisons awaiting measurements.** (a) Search progress under equal total model-and-tool cost at frozen prices; locked ID/OOD checkpoint assessment occurs after search and never guides it. (b) Final agreement under equal human budgets and a fixed evolution budget. No curves or gains are supplied. Table 11 specifies the controls.

before selection. The metric ensemble uses fixed normalized weights declared before human labels are inspected; fitting these weights belongs to the score-calibration arm. The static model judge uses a fixed prompt and no added tools, while the static tool judge is the complete unevolved seed. Score calibration and evaluator fine-tuning follow Appendix G. Evaluator fine-tuning changes model weights; score calibration changes numerical mappings over frozen seed signals. Prompt optimization uses the history-only allocation defined below, restricted to instructions in (K, V, G) with adapters, tools, and routes fixed. Program search uses that allocation with the same full editable scope as ERA. These internal controls do not stand for all implementations of GEPA, SkillOpt, or other published optimizers. Exact model/checkpoint versions, metric sets and normalizations, tuning configurations, and monetary caps remain pending in Appendix M and must be frozen before runs.

Search allocation contract. All four conditions use the same initial program, proposal-Train pool, agent model, edit interface, permitted diagnostics, acceptance guard, and terminal selection rule. Independent proposals always edit the current best program and receive no previous failed-attempt diagnostics. History-only proposals also edit the current best program, but can inspect their own run’s accumulated diagnostic history under the same access rules as ERA, without an active direction receiving priority or an unaccepted program becoming the parent. Depth-first allocation keeps the active direction and the program-inheritance rule of Section 5.3, without returning to a paused direction. The pause/resume extension additionally allows reopening a saved direction and its history, subject to the same eligibility rules. Its reopening policy remains to be frozen, and the row is design-only. The history-only versus depth-first contrast tests the combined allocation and inheritance policy, not either mechanism in isolation. Attributing a gain specifically to inheritance would additionally require depth-first allocation with the parent reset to the best program after every rejection.

The primary search budget is total model-and-tool cost evaluated under one frozen price schedule. It includes proposal tokens, requested diagnostics, failed calls, candidate measurement, and a reserved terminal-assessment allowance. All arms stop at the same cap; differences in numbers of attempts are outcomes, not a second matching requirement. Token counts, tool calls, elapsed time, and attempts are reported separately. Fixed post-search Test/OOD evaluation costs are also reported, but never buy additional search or select checkpoints. Fine-tuning requires a separate accounting of training compute; a common monetary cap alone does not make unavailable checkpoint or observation access comparable.

The comparisons in Table 11 hold nonintervened components and evidence access fixed under the declared resource contract. If method interfaces differ, report that difference instead of attributing an advantage to search allocation. Component attribution uses paired original-input groups and prespecified compatible replacements; interactions may prevent an independent per-component interpretation.

Table 11: Planned ablation conditions and the inference each comparison supports. All conditions require matched evidence access and accounted computation. An oracle condition is diagnostic, not a deployable comparator.

Factor	Conditions	Endpoint or interpretation
Data mining	Random; judge uncertainty; cross-model conflict; run instability; IPM (semantic multi-source disagreement)	Final held-out alignment at equal $N_H \in \{20, 50, 100, 200\}$; the number of selected disagreements alone is insufficient.
Human detail	Same pairs with $H_0, H_0 + H_1, H_0 + H_1 + H_2$, and $H_0 + H_1 + H_2 + H_3$	Marginal value of granularity; report annotation units and annotation time.
Editable scope	Full (A, K, V, T, Π, G) ; rubric/procedure only (K, V) ; fixed tools and static routes; routing only	Whether additional observations or structural changes are needed beyond criteria changes.
Evidence ceiling	Oracle evidence and/or oracle routing with access declared	Diagnostic headroom under privileged information; no deployment claim.
Search allocation	Independent one-shot proposals; history-only proposals; direction depth-first; direction search with pause/resume	Value of direction management at equal total search and measurement budgets.
Tool activation	Tool on/off with compatible consumers	Contribution of acquiring the observation, not the whole program bundle.
Tool implementation	Evolved tool versus seed tool	Contribution of changed measurement or evidence extraction.
Evidence route	Adaptive versus fixed routing with the same tool set	Contribution of which observations reach which modules.
Rubric and modules	Evolved versus fixed criteria/procedures	Contribution of judgment changes with observations held fixed.
Fusion	Declared numerical rule versus semantic synthesis	Contribution of resolving priorities, overlap, and conflicting findings.
Feedback adapter	Direct pairs versus derived ranking/pointwise/error labels on the same subset	Label fidelity and its downstream effect; keep original input dependence.

Table 12: Matched-budget search comparisons: unmeasured placeholders. All strategies share feedback, seed, editable components, and measurement protocol. Report independent-search means, dispersion, and run counts separately from input-group intervals for each frozen evaluator; account for all diagnostic and failed calls. The pause/resume row is a planned condition, not a claim of completed implementation or evaluation.

Search allocation	ID	OOD	Attempts/direction	Total cost
Independent proposals	—	—	—	—
History-only proposals	—	—	—	—
Direction depth-first	—	—	—	—
Depth-first + pause/resume	—	—	—	—

Table 12 contrasts history access, the combined depth-first policy, and the planned addition of pause/resume. The trajectory analysis records the cost before the first gain, later best-so-far gains, and the fate of each direction. Frozen-checkpoint ID/OOD assessment occurs only after search; it cannot select the next attempt. Independent searches, rather than repeated scoring of one final evaluator, quantify optimizer variability.

Existing-dataset mining replay. Existing output groups remain available to the miner, but their human labels are hidden until the sampling policy selects them. Freeze source mappings and the random anchor before targeted selection; no final label or mining-conflict score enters the deployed evaluator. Use the same original-input annotation units, feedback detail, and evolution budget for every policy. Replay tests label efficiency, not newly incurred human effort. Use historical annotation time only when compatible time records exist; otherwise record it as unavailable. For new

annotation, report agreement against both units and measured human time, including extra raters, adjudication, and abstentions. The random arm’s anchor is part of its uniformly sampled budget, not an additional set of labels.

K TRAJECTORY, MECHANISM, AND EFFICIENCY ANALYSIS

From unsuccessful implementation to accepted change. For each selected trajectory, report the direction’s initial error, hypothesis, candidate sequence, diagnostic measurements, and acceptance and inheritance decisions. A favorable example must show what changed between the failed and successful attempts, not merely that the later score is higher. Include unsuccessful and exhausted directions under the same selection rule. Judging a direction unhelpful requires a contradicted prediction and checks of relevant implementation problems; the controller does not make this judgment automatically.

Controlled component attribution. Compare compatible tool, route, rubric, module, and synthesis variants on the same inputs. Hold upstream observations fixed when testing their interpretation; rerun the full program when the observation path changes. A tool can matter only if its output reaches a consumer and changes the resulting judgment. Interactions between components prevent interpreting full-minus-fixed effects as additive contributions. Only after these tests should a failure be assigned to missing evidence, incorrect interpretation, or insufficient feedback.

Feedback and adapter fidelity. Assess H_0 through H_3 cumulatively on the same pairs and include annotation time. Direct and converted preferences are compared on the same original-input subset, first for label fidelity and then for downstream agreement. A label conversion does not create missing human rationales or dimensional judgments. Agreement among existing signals and agreement among annotators are distinct measurements.

Efficiency and deployment. Report data-mining computation, human time, search cost, and deployment cost separately. Search cost includes rejected candidates, diagnostic calls, retries, and terminal assessment. Repeated pointwise scoring measures execution variation; independent search seeds measure optimizer variation. A cost–agreement profile is not an evolution trajectory, and a longer direction does not by itself establish efficient search.

L SIMULATED REPORTING FIXTURES

Main-text C1 layout. Figure 2 retains the landscape layout but uses all 3,520 requested OOD input groups in the existing synthetic records, not the 2,880-input subset in Figure 5. Within each task, pairwise agreement uses the intersection of complete inputs across the five displayed methods; all five bars therefore share the same denominator. The intervals resample original inputs jointly across methods with 2,000 bootstrap draws, rather than treating pairs or scoring repeats as independent observations. Panel (b) averages these task means with equal weights within each family. Its selection endpoint uses all 160 requested inputs per task, counting failed scoring as failed Best-of-4 selection. The different endpoint denominators are recorded in `data/simulated/c1_landscape/`; no task is selected or omitted according to its outcomes. Connecting seed and ERA points identifies the same family, not a search trajectory or a causal effect. The four equal ring quadrants represent the task taxonomy, not measured sample proportions. The figure does not simulate calibration, fine-tuning, sampling efficiency, search progress, C2, or C3. The main table retains the full comparator specification, and no empirical claim is supported by this layout fixture.

Purpose and limits. The quantitative figures in this appendix and Table 14 are generated from explicitly simulated data for reviewing the reporting design. The generator does not execute ERA, run a model, consult benchmark outcomes, or recruit annotators. Its assigned noise, effects, costs, and cohort labels are not estimates of expected performance, a power analysis, or evidence for method ranking. Positive, near-zero, and negative contrasts are retained. None of these fixtures provides evidence for sampling efficiency, component effectiveness, or human-label reliability. No C3 training outcomes are simulated.

Table 13: Eight diagnostic cohorts for the simulated C2, annotation, and component displays. The C1 displays cover all 22 domains separately. Assignments are layout examples, not verified data availability or fixed domain categories. Each C1 domain/split contains 160 synthetic input groups and four outputs.

Family	Reused-preference example	New-preference example
Structured visuals	TD-SV4: specification-conditioned web	TD-SV2: webpage reconstruction
Images / 3D	TD-IG1: detailed image captioning	TD-IG3: instruction-based image editing
Text generation	TD-TG1: document summarization	TD-TG2: RAG-supported answers
Automated research	TD-AR1: research ideas	TD-AR2: ablation plans

Construction and uncertainty. The C1 fixtures cover all 22 domains in Table 2, with 11 illustrative cohorts per feedback setting. Table 13 retains eight fixed tasks for C2 and diagnostic detail; it does not define the study’s scope. A domain may use both feedback settings in an actual study. Gaussian latent utilities define synthetic C1 rankings. Noisy scores are repeated three times per artifact and reduced to medians before strict comparison. Incomplete groups remain in coverage denominators. The C1 table averages domain-level metrics equally within each setting. Its means use each method’s complete groups, whereas the paired differences in Figure 6 use the intersection of completed groups. Intervals use 1,000 input-group bootstrap replicates of these artificial observations. They quantify resampling variation in the fixtures, not uncertainty about ERA. The sampling-fixture generator assigns budget responses for five policies and 12 toy runs; its bands are percentile ranges across runs, not bootstrap intervals.

Balanced subset in the landscape figure. Figure 5 uses a subset of the OOD fixtures, with 720 input groups per family. The six-task families contain 120 inputs per task; the five-task families contain 144. Within each task, input identifiers are ranked by SHA256 with fixed seed 20260918, without reference to outcomes. The same selected inputs, including incomplete responses, are retained for every method. Agreement and its 1,000-replicate bootstrap intervals are recomputed on each method’s complete inputs; cost retains all selected inputs. Setting means weight their 11 domains equally, rather than pooling inputs. The ring’s equal quarters reflect this synthetic allocation, not measured domain prevalence or a proposed empirical sampling rule. The released selection list and numerical exports distinguish this display from the full-cohort tables and appendix plots.

Expanded alignment and sampling displays. Figure 6a–b reports per-domain ERA-minus-seed effects with paired intervals for all 22 domains on the full synthetic cohorts. These paired contrasts complement the marginal means for the balanced OOD subset in Figure 5c. Figure 7a–b compares sampling policies against random sampling at equal human budgets; all policies include the same reliability anchor. Annotation granularity is held fixed here and varied separately in Figure 10b.

Annotation outcomes. The annotation fixture assigns three simulated raters per group. Preferences for the two outputs are encoded as 0/1, a tie as 2, and abstention as -1 . Agreement is computed within each group from all available non-abstaining rater pairs, then averaged across eligible groups. Fewer than two valid votes is insufficient; three distinct valid votes yields no consensus. A two-of-three majority gives a decisive preference or a tie. These outcomes are not evidence about actual annotators or any public benchmark.

Selection beyond pairwise agreement. Figure 9a–b reports the entire selected-rank distribution. Rank 1 is the synthetic human optimum; unscored inputs remain visible. This full-input denominator differs from the completed-group Best-of-4 accuracy in Table 14. Both definitions are exported with their denominators so missingness cannot masquerade as better selection.

Mechanisms and feedback detail. The heatmap in Figure 10a specifies conditional contrasts between a full program and a program holding one component fixed. Such contrasts are not additive causal contributions. Panel (b) varies cumulative feedback on the same synthetic input groups. Its distributions describe assigned group-level values rather than independent search runs. Actual granularity studies must also account for annotation time and unequal label availability.

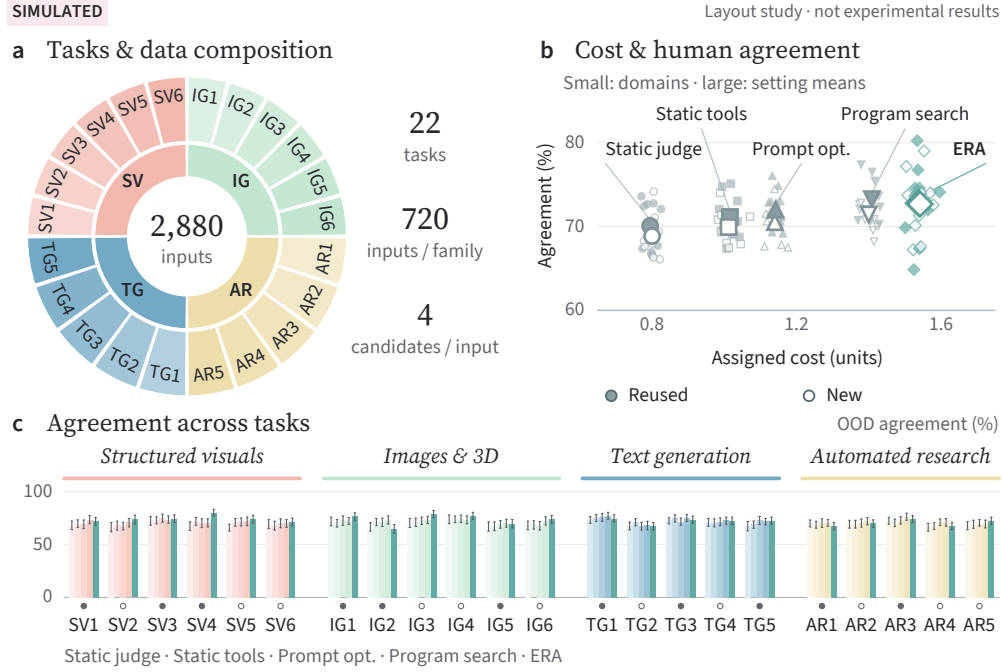


Figure 5: Simulated task coverage and evaluator comparisons; not experimental results. Panel (a) summarizes 22 tasks and 2,880 input groups, balanced to 720 per family with four outputs per input. A fixed hash selects this subset without reading outcomes; the appendix reports the full synthetic cohorts. Sector angles encode input counts; task codes follow Table 2. Panel (b) plots assigned cost units, not measured runtime, against OOD agreement with synthetic human rankings. Small marks denote 110 domain–method combinations; large marks equally average 11 domains per feedback setting. Shapes identify methods, and filled/open marks denote illustrative reused/new preferences, not label reliability. Panel (c) reports agreement from pointwise scores and 95% input-group bootstrap intervals on a common 0–100% scale. Baseline bars follow the displayed order in each family’s hue; ERA is lake green. Cost includes all selected inputs; agreement uses method-complete inputs. Marginal intervals do not establish paired differences.

SIMULATED DATA

NOT EXPERIMENTAL RESULTS

22 DOMAINS / 4 FAMILIES

Structured visuals · 6

SV1	Brief → single slide	R
SV2	Screenshot → webpage	N
SV3	Data → visualization	R
SV4	Specification → webpage	R
SV5	Sources → slide deck	N
SV6	Topic → slide deck	N

Images / 3D · 6

IG1	Detailed captioning	R
IG2	Text → image	R
IG3	Instruction-based editing	N
IG4	Super-resolution	N
IG5	Text → 3D	R
IG6	Image → 3D	N

Text generation · 5

TG1	Document summarization	R
TG2	RAG long-form answers	N
TG3	Translation	R
TG4	Instruction constraints	N
TG5	Open dialogue	R

Automated research · 5

AR1	Research ideas	R
AR2	Ablation plans	N
AR3	Table → insight	R
AR4	Related work	N
AR5	Peer-review critique	N

a ID test

b OOD test

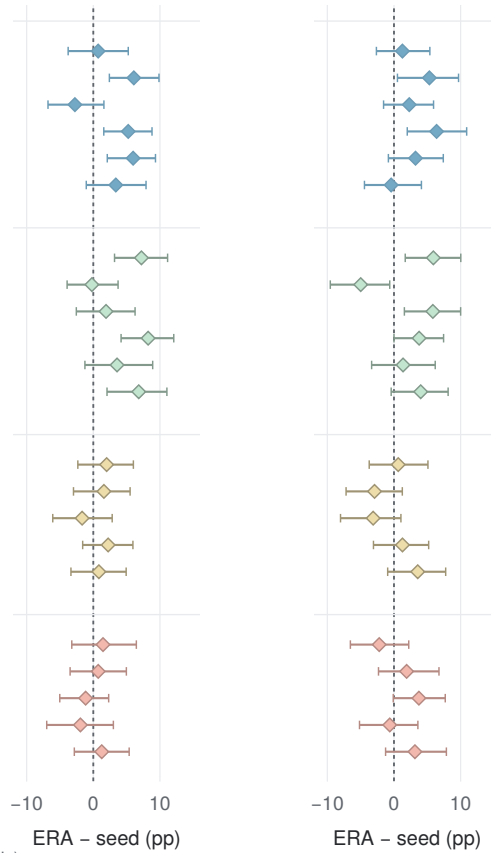


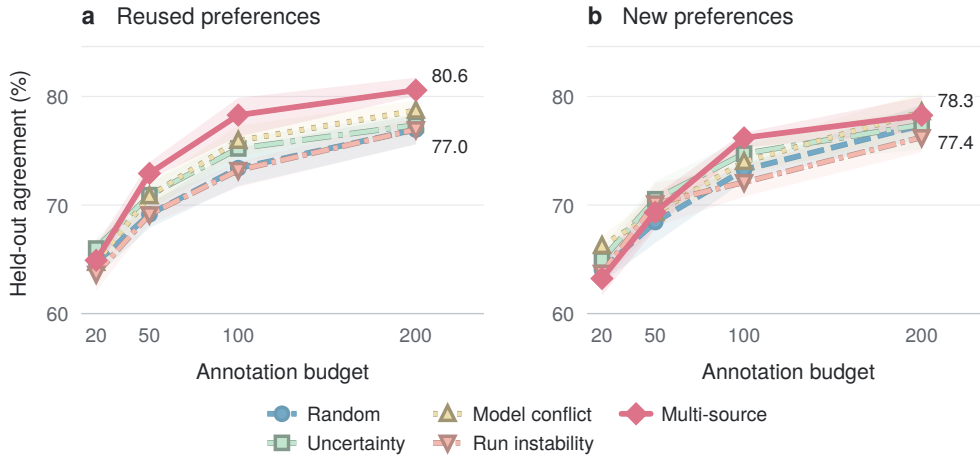
Figure 6: **Simulated domain-level alignment; not experimental results.** Panels (a) and (b) compare ERA with the static tool-augmented seed on ID and OOD inputs. Diamonds are paired mean differences; whiskers are 95% bootstrap intervals over jointly completed synthetic input groups. Zero denotes no difference. All 22 domains retain the 6/6/5/5 taxonomy order. Colors identify domain families; R/N denote illustrative reused/new feedback assignments, not permanent domain categories or measured data readiness.

Table 14: **Simulated C1 reporting example; not experimental results.** Each setting averages eleven equally weighted toy cohorts. Pair = pairwise agreement; Top = top-region agreement; Bo4 = Best-of-4 accuracy (all in %); Reg = normalized rank regret (lower is better). Cohort assignments do not assert actual label availability. Coverage and domain summaries accompany the reproducible fixtures.

Evaluator	ID test				OOD test			
	Pair	Top	Bo4	Reg	Pair	Top	Bo4	Reg
SIMULATED Reused preferences								
Single metric	66.5	67.5	44.8	0.31	64.1	65.2	43.5	0.32
Metric ensemble	69.9	70.9	51.2	0.26	66.8	68.0	45.3	0.30
Static judge	71.3	72.3	52.2	0.25	69.9	71.1	49.8	0.27
Static tools	74.0	75.2	57.1	0.23	71.3	72.9	52.8	0.24
Prompt optimization	74.1	75.5	56.8	0.22	71.9	72.9	54.4	0.24
Program search	74.8	76.1	57.7	0.21	73.0	74.5	55.5	0.23
ERA	75.3	76.9	61.2	0.20	72.7	74.2	55.3	0.23
SIMULATED New preferences								
Single metric	66.2	67.0	45.1	0.30	63.3	64.3	40.5	0.34
Metric ensemble	68.3	69.6	47.8	0.29	65.9	66.9	43.8	0.32
Static judge	71.3	72.5	53.5	0.24	68.4	69.6	48.0	0.29
Static tools	72.4	73.6	54.9	0.24	70.0	71.2	51.3	0.26
Prompt optimization	74.3	75.5	56.8	0.22	70.6	71.8	51.3	0.26
Program search	74.3	75.7	56.5	0.22	71.6	72.9	53.4	0.24
ERA	75.8	77.1	59.9	0.20	72.2	73.6	53.4	0.24

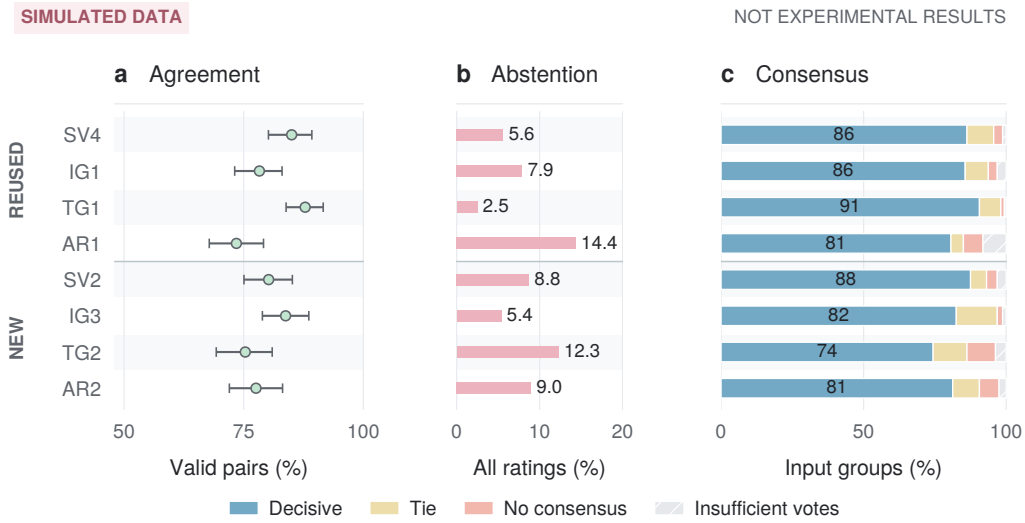
SIMULATED DATA

NOT EXPERIMENTAL RESULTS



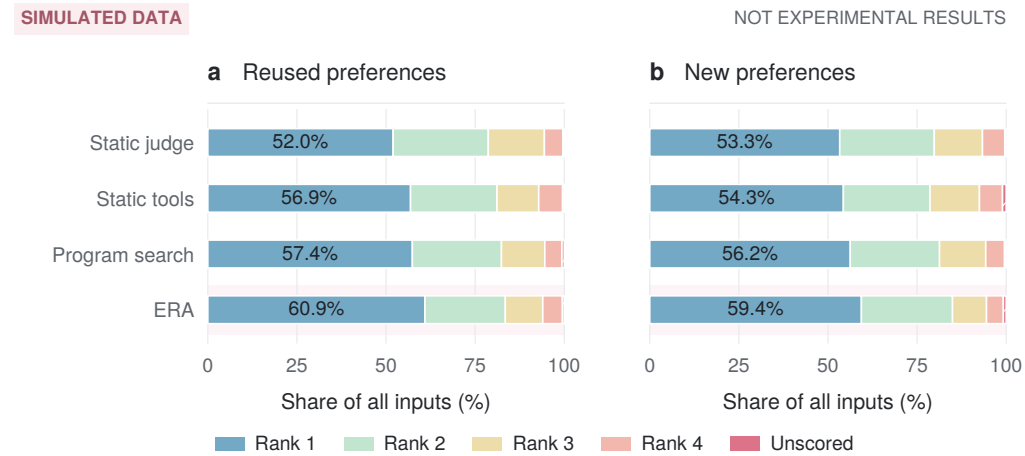
Bands: 10–90% of 12 toy runs. Each budget includes the same 10-unit random anchor.

Figure 7: **Simulated sampling curves; not experimental results.** Panels (a) and (b) compare five sampling policies under reused and newly collected preferences. Lines average 12 toy runs; bands mark each policy’s 10th–90th percentiles, not confidence intervals. Budgets count original-input annotation units, including a shared 10-unit random anchor.



Three simulated raters per group. Agreement, abstention, and consensus use distinct denominators.

Figure 8: **Simulated annotation audit; no human study was conducted.** Panel (a) averages valid-pair agreement within input groups; whiskers are 95% group-bootstrap intervals. Panel (b) divides abstentions by all requested ratings. Panel (c) retains decisive preferences, ties, no consensus, and insufficient votes under a two-of-three rule. Domain codes map to Table 13. The settings share a schema, not an assumed reliability ordering.



ID test; equal-sized toy cohorts. Rank 1 is best. Failed scoring remains in the denominator.

Figure 9: **Simulated C1 selection; not experimental results.** Panels (a) and (b) show horizontal rank profiles for four evaluator fixtures in the two construction settings, using ID inputs. Rank 1 is best. Labels give its share of all input groups, including unscored groups in the denominator. Colors encode ranks, not methods. Equal cohort sizes permit pooling within a setting; actual domain-level results must also be reported.

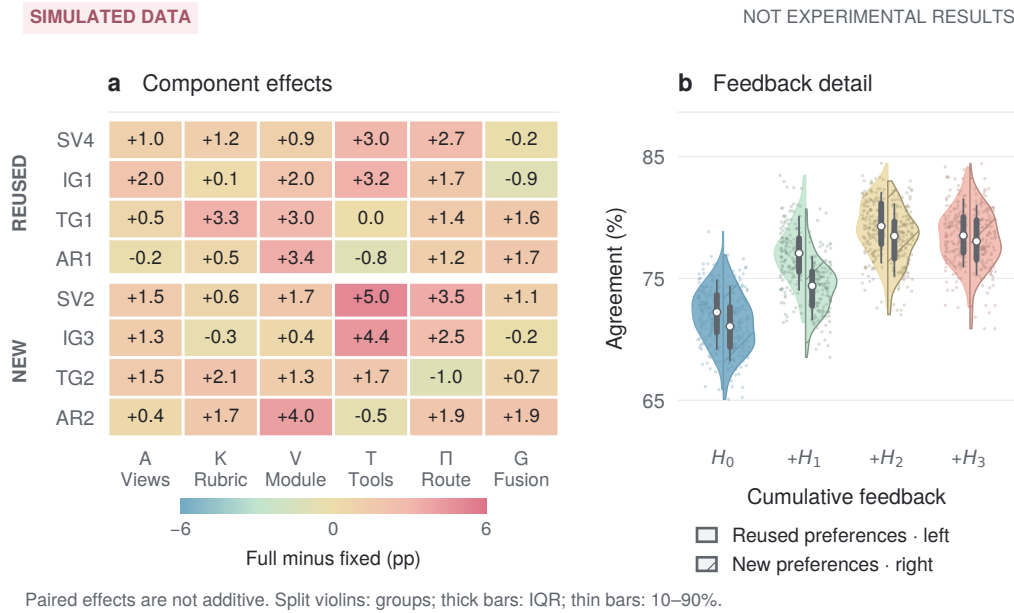
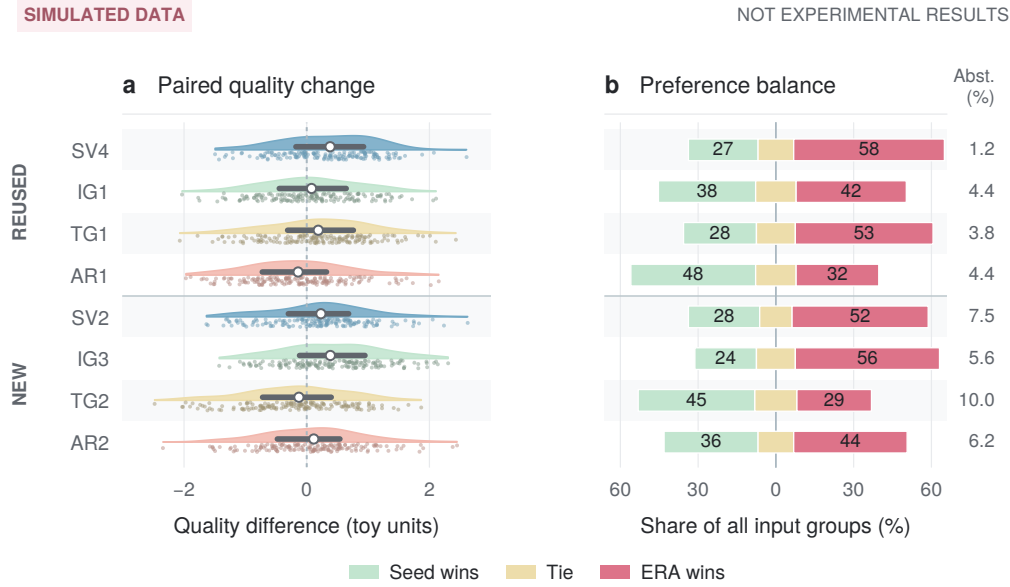


Figure 10: **Simulated ablations; not experimental results.** Panel (a) gives mean full-minus-fixed agreement (percentage points) for (A, K, V, T, Π, G) across eight diagnostic domains, with cream at zero. Panel (b) colors cumulative H_0 – H_3 levels. Left/right halves denote reused/new preferences; right halves are hatched. Dots are synthetic groups, white markers medians, thick segments quartiles, and thin segments the 10th–90th percentiles. Domain codes follow Table 13. Conditional contrasts are not additive contributions or evidence of component benefits.

Refinement and resource use. The C2 fixture constructs two noisy refinements from a shared latent y_0 . Figure 11a retains their paired quality differences and panel (b) converts those differences to wins, ties, and losses with an independent abstention indicator. The toy tie threshold is 0.15 latent units, not a proposed universal human threshold. Reused/new row labels describe feedback used for evaluator evolution; endpoint judgments on fresh C2 outputs must be new in both settings. Changes against y_0 and major-regression flags are retained in the data for the full reporting contract.

Figure 12a–b aligns assigned relative deployment costs and agreement on common evaluator rows, keeping both settings visible. Connecting segments pair settings, not steps of evolution. A real resource comparison must report search and annotation costs separately from deployment cost; the artificial costs here cannot establish practical efficiency.

Reproduction and replacement. The offline script `scripts/render_simulated_results.py`, seeded with 20260916, produces the CSV fixtures, vector figures, numerical table, and a manifest of hashes. Every data row carries a `SIMULATED` provenance field, and every chart discloses its status within the image. The committed assets compile without model access. Before a results-bearing submission, replace the fixtures with authorized measurements, regenerate captions and uncertainty, and audit all claims against the actual evidence. Removing a disclosure without replacing its data is not a valid conversion to results.



Difference = ERA-refined minus Seed-refined. All artifacts and judgments here are synthetic.

Figure 11: **Simulated C2 outcomes; not experimental results.** Panel (a) combines half-violin densities, all synthetic group values, median markers, and interquartile segments for ERA-refined minus Seed-refined quality. Density colors identify domain families. The horizontal scale is an arbitrary latent unit, not a human score. Panel (b) centers ties between seed and ERA wins, with abstentions in a separate percentage column. Every share uses all input groups under the toy rule. Both construction settings would require fresh blinded human endpoint judgments in an actual C2 study.



Lines pair the two settings for one evaluator. Costs are assigned, not measured compute.

Figure 12: **Simulated quality-cost trade-offs; not measured costs.** Panel (a) gives assigned costs and panel (b) gives agreement, each averaging eleven ID cohorts equally per setting. Colors identify evaluators; metric-only references are gray. Filled circles denote reused preferences and hollow squares new preferences; connecting segments pair the settings for one evaluator. Cost units are assigned to the toy generator, not timed API calls or measured compute. The plot does not establish an efficiency frontier for ERA.

M EXPERIMENT MANIFEST

Table 15: Information required before confirmatory evaluation. Unset values are intentional placeholders, not inferred development defaults.

Item	Required specification
Domain and sources	Versioned taxonomy, decision rule, readiness audit, dataset versions, permissions, input counts, output groups: <i>TBD</i>
Human feedback	Direct/derived provenance, H_0-H_3 availability, core dimensions, annotator pool, agreement, time, tie policy: <i>TBD</i>
Partitioning	Original-input/source grouping, deduplication, split sizes, OOD definition, External role, contamination review: <i>TBD</i>
Mining	Source inventory, mappings, dependence groups, anchor size, calibration, strata, frozen policy, human-budget accounting: <i>TBD</i>
Models	Frozen scorer, proposer, tool-model, and generator identities, revisions, decoding, and permissible evidence: <i>TBD</i>
Program and search	Seed, editable components, edit bounds, allocation, development guard, protection criteria, shortlist rule: <i>TBD</i>
Measurement	Repeats $R \geq 3$, precision, completeness, retries, artifact reuse, cache identity, search seeds, stopping rule: <i>TBD</i>
Budgets	Human time, opportunities, model calls/tokens, tools, runtime, deployment cost, training compute, matching rule: <i>TBD</i>
Downstream controls	C1 pool size; C2 fresh generation and refinement; C3 reward, LoRA/GRPO settings, pilot gates and preflight: <i>TBD</i>
Analysis	Primary contrasts, grouped intervals, multiplicity, missingness, reserved metrics, minimum effects, final sample sizes: <i>TBD</i>

Before submission, complete Table 15, replace result placeholders with permitted source-traceable measurements, finalize the empirical scope without changing the method’s definition of a domain, and supply appropriately anonymized reproducibility artifacts. Unmeasured hypotheses remain hypotheses.